



Radial gradient of superionic hydrogen in the Earth's inner core

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Hydrogen is considered a key light element in Earth's core, yet the thermodynamics of its superionic phase and its distribution in the inner core remain unclear. Here, we compute *ab initio* Gibbs free energies for liquid and superionic hcp and bcc Fe–H phases and construct the superionic–liquid phase diagram over pressure–temperature conditions relevant to the Earth's inner core. We find that phase diagrams at different inner-core pressures collapse when temperatures are scaled by the melting temperature of pure iron, indicating that solid–liquid partitioning is controlled primarily by a reduced temperature relative to iron melting and is weakly sensitive to pressure. This scaling relation further reconciles previously reported discrepancies in partition coefficients among theoretical studies and yields good agreement with available experimental data at low pressures. By applying thermochemical constraints, our free-energy results reveal a radial hydrogen gradient within the inner core. These results demonstrate that compositional gradients of superionic hydrogen in the inner core emerge naturally from equilibrium thermodynamics and suggest a general mechanism governing the depth-dependent distribution of light elements within Earth's inner core.

Earth's core | superionic state | phase diagram

The Earth's core consists of a solid inner core (IC) surrounded by a liquid outer core (OC). Both regions are dominated by iron, but additional light elements are required to explain the core's density deficit compared to pure Fe (1–4). The composition and distribution of these light elements are therefore central to understanding the structure and evolution of the core (1, 5, 6). While the OC is generally found to be well mixed, the IC exhibits a more complex internal structure, including pronounced seismic anisotropy and heterogeneities (5–7). These anomalous seismic features have been attributed to the elastic properties of core-forming materials and preferred lattice orientations in the IC (6, 8, 9). Based on comparisons of the elastic properties of Fe alloys with different compositions, recent experimental studies have further suggested that radial compositional gradients within the IC may provide improved explanations for the observed seismic anisotropy (10, 11). However, the origin of these gradients remains poorly understood.

Hydrogen is considered a key light element in the core (1, 3, 5), owing to its high cosmic abundance and strong siderophile behavior under high pressure–temperature conditions (12–14). Recent work has suggested that hydrogen enters a superionic state in the hexagonal close-packed (hcp) structure of Fe under IC conditions, exhibiting liquid-like diffusion within the crystalline lattice (15, 16). The superionic state is fundamentally distinct from a conventional solid and can affect the thermodynamic properties of different phases (17). Although the IC is commonly assumed to adopt the hcp structure (18, 19), several elements have been shown to stabilize the body-centered cubic (bcc) phase at core conditions (20–24). Whether hydrogen can also exist as a superionic species in the bcc lattice, and how it influences the competition between hcp and bcc phases in the IC, remains unclear.

Given the potentially large abundance of hydrogen in the core, its partitioning between solid and liquid iron has been extensively investigated in order to constrain hydrogen concentrations in both the IC and OC. High-pressure experiments by Oka et al. and Hikosaka et al. reported solid–liquid partition coefficients of hydrogen, $D_{\text{H}}^{\text{S/L}}$, of approximately 0.7 to 0.8 at (1,900 K, 45 GPa) (25) and (2,100 K, 56 GPa) (26), respectively. Because direct experiments at inner-core boundary (ICB) conditions remain challenging, recent theoretical studies have used machine-learning interatomic potentials (MLPs) to estimate $D_{\text{H}}^{\text{S/L}}$ at higher pressures and temperatures. These studies, however, report largely varying results: Yuan and Steinle-Neumann (27) obtained values of 0.50 to 0.62 under ICB conditions, Liu and Jing (28) reported smaller values of 0.29 to 0.46 at ICB pressure, and Zhang et al. (29) suggested a nearly constant value of ~0.56 across IC conditions. The origin of these discrepancies is unclear, as these simulations employed different MLPs

Significance

Earth's inner core is composed mainly of iron with small amounts of light elements. Hydrogen is a key candidate because of its high cosmic abundance and strong affinity for iron. Under inner-core conditions, hydrogen adopts a superionic state, exhibiting liquid-like mobility within a crystalline lattice. This work computes complete superionic–liquid phase diagrams over a range of core pressures and temperatures and shows that they obey a simple thermodynamic scaling law. This scaling reconciles previously inconsistent theoretical estimates of hydrogen partitioning between solid and liquid iron and brings them into agreement with low-pressure experiments. Applying these constraints reveals a radial hydrogen gradient in the inner core, providing a unified framework for the distribution of light elements in Earth's core.

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and different protocols for computing the partition coefficient. Such inconsistencies hinder further development of compositional models for multicomponent Fe–light-element alloys in the core.

Both element partitioning and phase competition are essentially governed by the thermodynamic properties of the phases. A comprehensive understanding of hydrogen in the IC therefore requires direct access to the Gibbs free energies of superionic and liquid Fe–H phases, from which phase diagrams and equilibrium compositions can be determined. In this work, we employ a recently developed ab initio framework to compute Gibbs free energies and construct phase diagrams for superionic and liquid Fe–H under core conditions (24). This framework has been demonstrated to accurately describe the superionic Fe–O system at extreme pressures and temperatures (24). Using this approach, we investigate the phase competition among superionic hcp, superionic bcc, and liquid Fe–H, assess the pressure dependence of the superionic–liquid phase diagram, and establish thermodynamic constraints on hydrogen distribution from ICB to the center of the IC. We aim to provide interpretations of previously reported discrepancies in $D_{\text{H}}^{\text{S/L}}$ and to elucidate the thermodynamic origin of a possible radial chemical gradient in the IC.

Results

Superionic–Liquid Phase Diagram. We first simulate Fe–H alloy in the hcp, bcc, as well as liquid phase with ab initio molecular dynamics (AIMD). Hydrogen exhibits a superionic state in both hcp and bcc lattices, where its diffusion coefficient is comparable to that in the liquid phase (SI Appendix, Fig. S1). Fig. 1A shows the partial pair correlation functions (PPCF) of $\text{Fe}_{85}\text{H}_{15}$ in superionic hcp, superionic bcc, and liquid phases at 323 GPa and 5,500 K. The Fe–H PPCF shows a nearest-neighbor peak at ~ 1.5 Å in all phases, indicating strong Fe–H bonding that is insensitive to the Fe phase. The H–H PPCF rapidly approaches unity, indicating weak H–H bonding in all three phases, consistent with the liquid-like, superionic behavior of hydrogen within the crystalline lattices.

To study the superionic–liquid equilibrium at a large scale, we develop an interatomic potential for Fe–H system under core conditions, capable of simulating the superionic state in both hcp and bcc phases (Materials and Methods and SI Appendix, Note S1). This potential enables superionic–liquid coexistence simulations (Fig. 1B), and classical-to-ab initio thermodynamic integration calculations (30) to obtain the ab initio Gibbs free energies of the superionic and liquid phases at 323 GPa over a wide range of

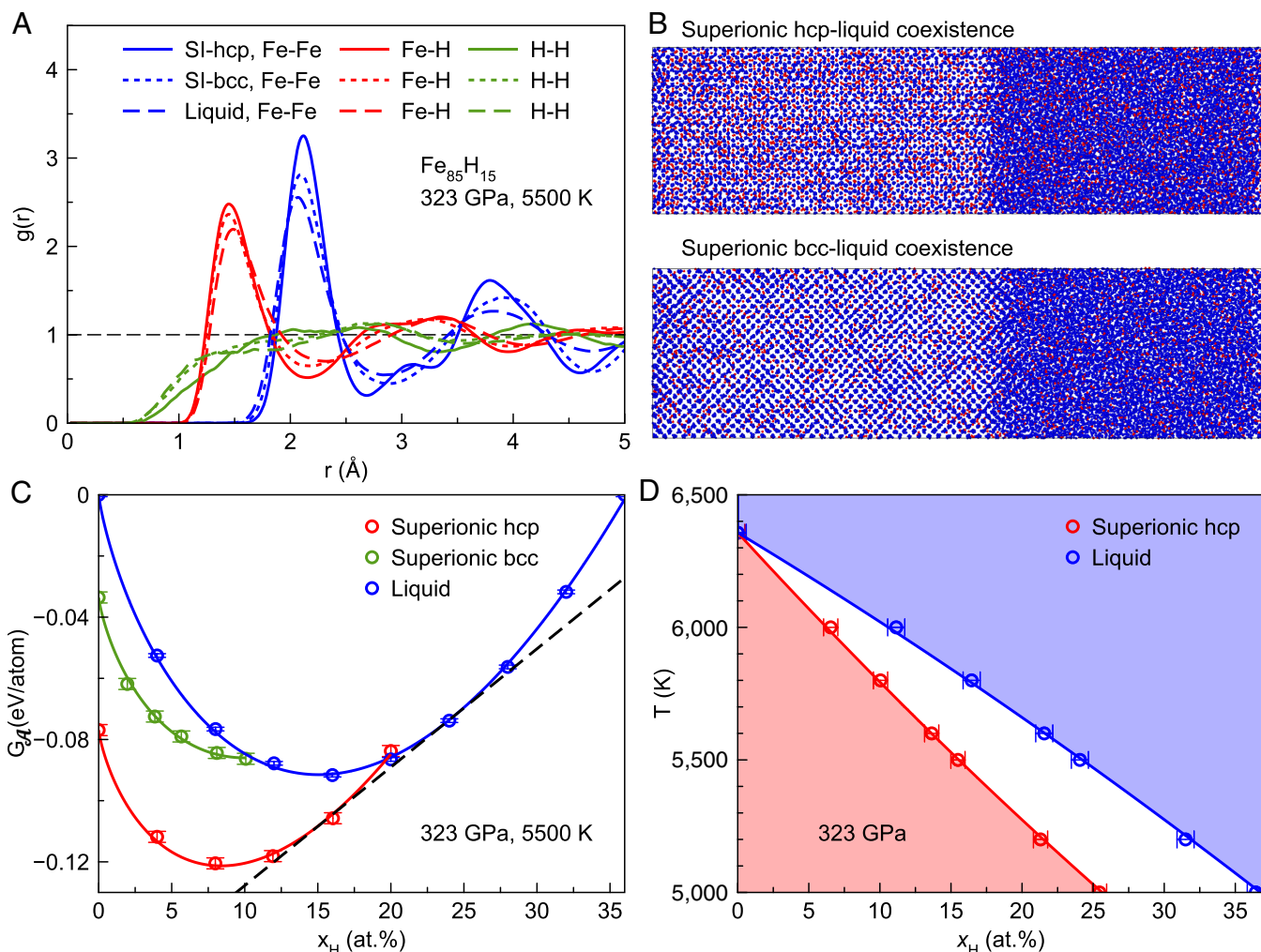


Fig. 1. Ab initio superionic–liquid phase diagrams of $\text{Fe}_{1-x}\text{H}_x$ at 323 GPa. (A) PPCF for superionic hcp, superionic bcc, and liquid $\text{Fe}_{85}\text{H}_{15}$ at 323 GPa and 5,500 K from AIMD. (B) Superionic hcp–liquid and bcc–liquid coexistence simulations. Blue and red dots represent iron and hydrogen atoms, respectively. (C) Ab initio Gibbs free energies of the liquid, superionic hcp, and superionic bcc $\text{Fe}_{1-x}\text{H}_x$ phases at 5,500 K and 323 GPa, referenced to the liquid free energy at $x_{\text{H}} = 0$ at.% and $x_{\text{H}} = 36$ at.% for clarity. The black dashed line denotes the common tangent between the liquid and superionic hcp free-energy curves. (D) Ab initio superionic–liquid phase diagrams of $\text{Fe}_{1-x}\text{H}_x$ at 323 GPa. The red and blue represent the superionic hcp-stable and liquid-stable regions, respectively.

temperatures (*SI Appendix, Note S2 and Figs. S3–S9*). The Gibbs free energies at 5,500 K for superionic hcp, superionic bcc, and the liquid are shown as an example in Fig. 1C. For all hydrogen compositions investigated here, the bcc phase consistently exhibits a higher free energy than the hcp phase. Thus, similar to pure Fe (19), the bcc phase remains metastable, while the hcp phase is thermodynamically stable with hydrogen contents up to ~20 at.%. Therefore, only the superionic hcp phase can coexist in equilibrium with the liquid in Fe–H system. The uncertainties in the free-energy calculations are 1 to 2 meV/atom for the superionic and liquid phases (*SI Appendix, Note S4*). We determine the solidus and liquidus curves by constructing the common tangent between the liquid and superionic hcp free-energy curves, as illustrated for 5,500 K in Fig. 1C. Repeating this procedure at different temperatures (*SI Appendix, Fig. S9*) yields the superionic–liquid phase diagram at 323 GPa shown in Fig. 1D. In contrast to the small solubility of superionic oxygen (~0.4 at.%) at 5,800 K and 323 GPa (24), hydrogen exhibits a much higher solubility, reaching 10 at.% under the same conditions.

We perform a large number of free-energy calculations for superionic hcp, superionic bcc, and liquid Fe–H phases with

different compositions to cover a broad pressure–temperature range relevant to Earth’s IC. Fig. 2A shows an example of the free energy calculation for superionic hcp with $x_H=20$ at.%. Starting from 6,000 K and 323 GPa, we compute the Gibbs free energy along the isothermal path using the thermodynamic integration of pressure–volume data at 6,000 K over 323 to 360 GPa obtained from AIMD simulations (*Materials and Methods and SI Appendix, Note S3 and Figs. S10–S15*). Then at 360 GPa, we employ the Gibbs–Helmholtz relation to compute the free energy along the isobaric path. In this manner, Gibbs free energies are computed over the full pressure–temperature space spanning 323 to 360 GPa and 5,000 to 6,800 K for superionic hcp, superionic bcc, and liquid phases of different compositions. The compositional dependence of these free energies is fitted using the Redlich–Kister (RK) expansion (31).

The resulting free-energy data enable analysis of phase competition over a wide range of P – T – x conditions. We first compare the free energies of the superionic hcp and bcc phases. While the superionic hcp phase is stable across most of the investigated P – T – x space, sufficiently high temperatures and hydrogen contents can stabilize the superionic bcc phase. For example, in Fig. 2B, the superionic bcc

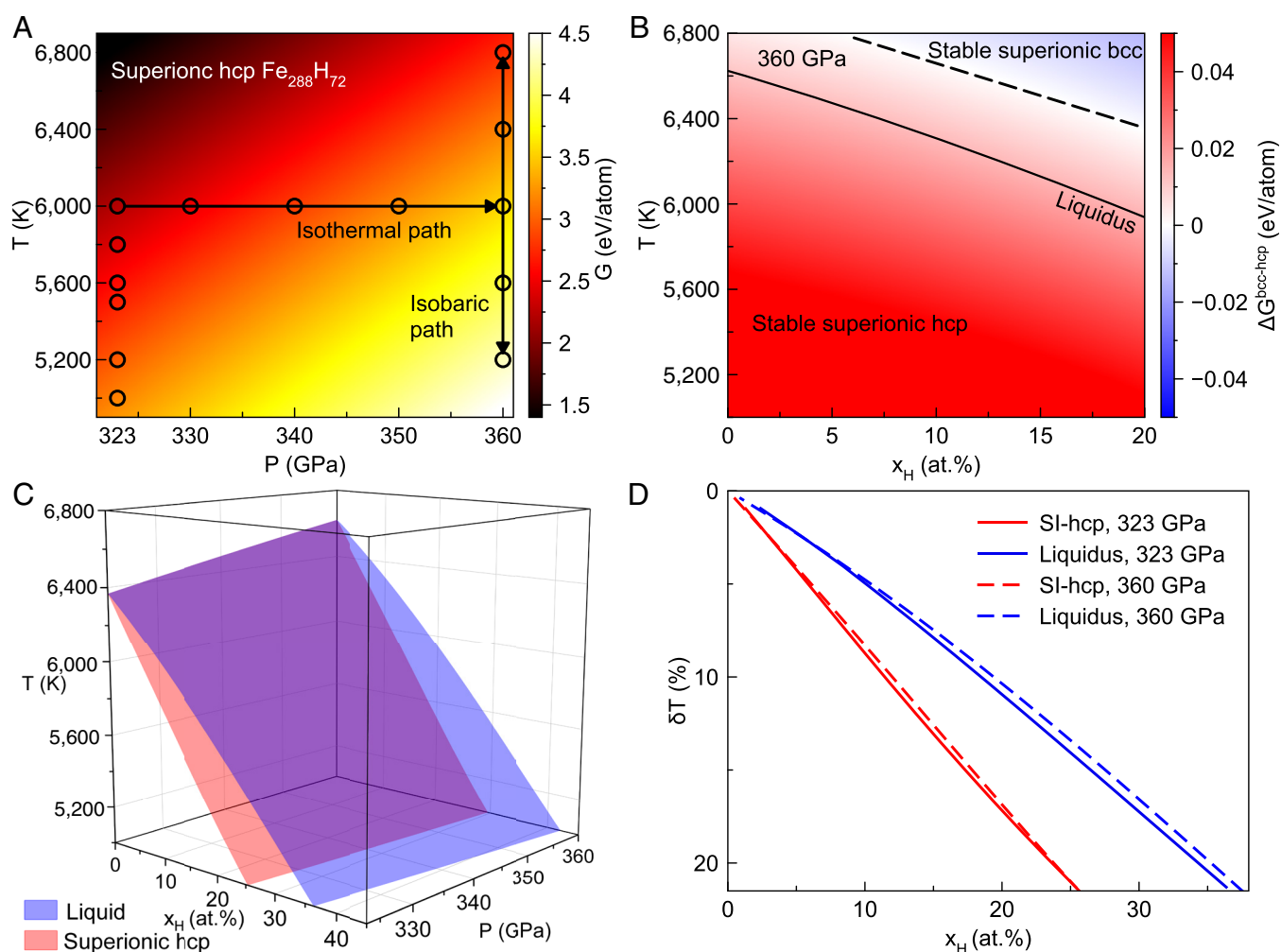


Fig. 2. Ab initio Gibbs free energies and phase diagram of $\text{Fe}_{1-x}\text{H}_x$ in the pressure range of 323 to 360 GPa. (A) Ab initio Gibbs free energies of superionic hcp $\text{Fe}_{288}\text{H}_{72}$ phase over 323 to 360 GPa and 5,000 to 6,800 K. Black open symbols represent P – T data used to calculate free energy along isothermal and isobaric paths using Eqs. 5 and 6. The color map shows the free-energy magnitude. (B) Gibbs free energy differences between superionic bcc and superionic hcp phases, $\Delta G^{\text{bcc-hcp}}(x, T)$, as functions of temperature and H content at 360 GPa. The color map represents the free-energy difference, with red and blue regions corresponding to the stability fields of the superionic hcp and bcc phases, respectively. The black dashed line indicates where $\Delta G^{\text{bcc-hcp}} = 0$. The solid line denotes the liquidus corresponding to coexistence between superionic hcp and liquid phases. (C) Ab initio superionic–liquid phase diagram of the Fe–H system from 323 to 360 GPa. Red and blue surfaces represent the solidus and liquidus, respectively, as functions of pressure. (D) Solidus and liquidus curves at 323 and 360 GPa plotted using the reduced temperature, δT , defined as $\delta T = (T_{m,\text{Fe}} - T)/T_{m,\text{Fe}}$, where $T_{m,\text{Fe}}$ is the iron’s melting temperature at the corresponding pressure.

phase becomes more stable than the hcp phase when the temperature exceeds 6,400 K and the hydrogen content exceeds 20 at.% at 360 GPa. However, when the liquid phase is taken into account, the stability field of superionic bcc is unlikely to be realized, as the hcp–bcc phase boundary lies above the liquidus line in Fig. 2B. Fig. 2C presents the superionic–liquid phase diagram between 323 and 360 GPa, where only the hcp phase appears as the stable solid. Both the solidus and liquidus lines shift systematically to higher temperatures with increasing pressure. In Fig. 2D, we find that when temperature is normalized by the melting temperature of pure Fe at each pressure, using the reduced temperature $\delta T = (T_{m,Fe} - T)/T_{m,Fe}$ the solidus and liquidus lines at 323 and 360 GPa collapse onto nearly identical curves. This scaling behavior indicates that pressure primarily rescales the absolute free-energy scale of pure Fe, while leaving the thermodynamics governing hydrogen mixing in superionic and liquid Fe phases largely unchanged.

Hydrogen Distribution in the IC. The superionic–liquid equilibrium condition at the ICB constrains the H composition in both the liquid (x_{ICB}^L) and superionic solid (x_{ICB}^{SI}), through equality of their chemical potentials (μ_H), i.e., $\mu_H^L(x_{ICB}^L, T_{ICB}, P_{ICB}) = \mu_H^{SI}(x_{ICB}^{SI}, T_{ICB}, P_{ICB})$. More generally, thermodynamic equilibrium also constrains the hydrogen distribution within the IC at different depths. Because hydrogen remains in a liquid-like superionic state throughout the IC, its chemical potential should be spatially uniform, such that

$$\mu_H^{SI}(x_H^{SI}(r_{IC}), T(r_{IC}), P(r_{IC})) = \mu_H^{SI}(x_{ICB}^{SI}, T_{ICB}, P_{ICB}), \quad [1]$$

where r_{IC} denotes the radial distance from the center of the IC. This condition provides a strong thermochemical constraint on the hydrogen distribution within the IC. Using the complete free-energy dataset obtained in this work, we explore the hydrogen distribution in the IC under these thermodynamic constraints. Based on the current understanding of the core's thermal structure (32), the IC is nearly isothermal due to its high thermal conductivity (33). We therefore examine the hydrogen chemical potential under isothermal conditions by setting $T(r_{IC}) = T_{ICB}$ and adopting $T_{ICB} = 5,500$ K (5). Fig. 3A shows the Gibbs free energy of superionic hcp Fe–H at 5,500 K over the pressure range from 330 to 365 GPa, spanning conditions relevant to the IC. We find these free-energy data are well fitted using the RK model with a fitting error of ~ 1 meV/atom. This model enables calculation of the hydrogen chemical potential (34) as

$$\mu_H^{SI}(x_H^{SI}) = G^{SI}(x_H^{SI}) + (1 - x_H^{SI}) \frac{\partial G^{SI}(x)}{\partial x}. \quad [2]$$

Fig. 3B shows μ_H^{SI} in the superionic hcp phase at 5,500 K across a range of pressures. For $T_{ICB} = 5,500$ K, the phase diagram at 330 GPa in Fig. 2C constrains the solidus x_{ICB}^{SI} and liquidus x_{ICB}^L . As a result, the hydrogen chemical potential in the IC is constrained, as indicated by the dashed line in Fig. 3B. This constraint leads to

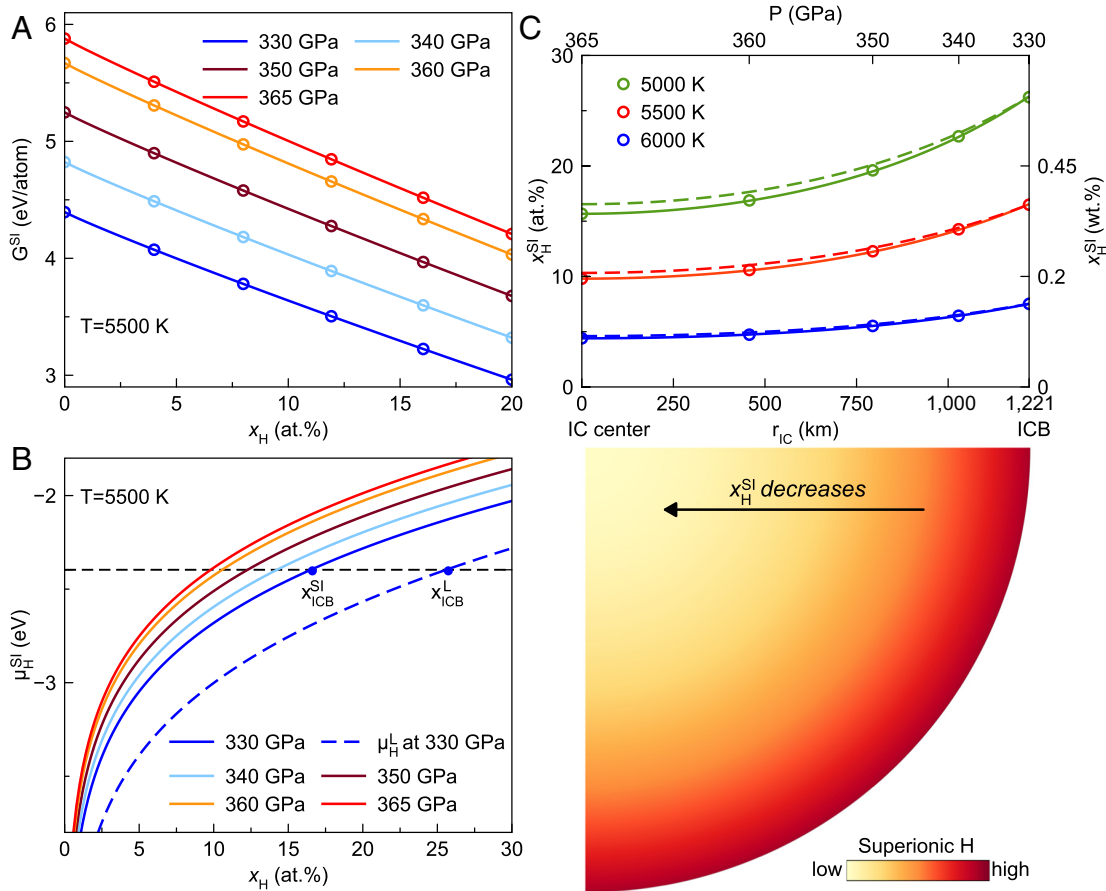


Fig. 3. Radial gradient of superionic hydrogen in Earth's inner core. (A) Gibbs free energy of superionic hcp Fe–H as a function of hydrogen content at 5,500 K for different pressures. The solid lines are fits to the data using Eq. 7. (B) Hydrogen chemical potential in the superionic hcp phase as a function of hydrogen content (solid curves). The dashed blue curve represents the hydrogen chemical potential in the liquid phase at 330 GPa and 5,500 K. The dots represent the hydrogen content constrained by the superionic–liquid equilibrium at the ICB under the conditions of 330 GPa and 5,500 K. The black dashed line represents a constant hydrogen chemical potential throughout the IC. (C) Top: Hydrogen concentration as a function of radial distance in the IC. The dashed lines represent the results including the effect of gravity. Bottom: Schematic illustration of the hydrogen concentration gradient within the IC. The color map shows the concentration of superionic hydrogen.

varying hydrogen compositions at different pressures, corresponding to different depths within the IC. The resulting depth-dependent hydrogen distribution is presented in Fig. 3C. At 5,500 K, the H composition is ~16 at.% at ICB ($r_{\text{IC}} = 1,221$ km). As the radius decreases, the H composition drops sharply, reaching about 12 at.% at $r_{\text{IC}} = 750$ km. Toward the IC center, the variation of H composition becomes more gradual. From $r_{\text{IC}} = 250$ km to the center, the H composition remains nearly constant. Overall, at 5,500 K, the hydrogen concentration decreases by ~7 at.% from the ICB to the IC center. We also perform similar analyses using $T_{\text{ICB}} = 5,000$ K and 6,000 K, respectively (SI Appendix, Fig. S16). In all cases, the IC exhibits compositional gradients, although the absolute gradients differ with T_{ICB} . At 5,000 K, the difference in hydrogen concentration between the ICB and the center of the IC is ~10 at.%, whereas at 6,000 K this difference decreases to ~3 at.%. We also assess the effect of the temperature profile on the chemical gradient by considering an adiabatic limit (SI Appendix, Fig. S17). Even under this upper-bound temperature gradient, a hydrogen concentration gradient persists throughout the IC.

Recently, Ganguly discussed the equilibrium radial distribution of light elements in the liquid OC, where the effect of gravity was also considered (35). Here, we evaluate the hydrogen distribution by explicitly accounting for the gravitational field at different depths, under the equilibrium condition:

$$\begin{aligned} \mu_{\text{H}}^{\text{SI}}(x_{\text{H}}^{\text{SI}}(r_{\text{IC}}), T(r_{\text{IC}}), P(r_{\text{IC}})) + \Phi(r_{\text{IC}}) \\ = \mu_{\text{H}}^{\text{SI}}(x_{\text{H}}^{\text{SI}}, T_{\text{ICB}}, P_{\text{ICB}}) + \Phi(r_{\text{ICB}}), \end{aligned} \quad [3]$$

where $\Phi(r_{\text{IC}})$ is the gravitational potential energy, given by $\Phi(r_{\text{IC}}) = \int_0^{r_{\text{IC}}} m_{\text{H}} g dr$. As shown in Fig. 3C, including gravitational potential energy slightly modifies the hydrogen composition, but the gradient in the IC remains unchanged. This is because the gravitational potential energy difference between the IC center and the ICB is significantly smaller than the corresponding variation in chemical potential (SI Appendix, Fig. S18).

Discussion

We have computed the free energies and phase diagram of superionic and liquid Fe–H phases over a broad range of pressure and temperature conditions relevant to the Earth's IC. To compare

with previous work, we calculate the molar partition coefficient between the superionic and liquid phases, $D_{\text{mole}}^{\text{S/L}}$, by taking the ratio of the solidus and liquidus compositions directly from the phase diagram in Fig. 2C. As shown in Fig. 4A, $D_{\text{mole}}^{\text{S/L}}$ ranges from 0.70 at 5,000 K to 0.59 at 6,000 K at 323 GPa, and from 0.68 at 5,200 K to 0.57 at 6,400 K at 360 GPa, exhibiting nearly linear dependence on temperature under both pressures. Previous simulations reported different values of $D_{\text{mole}}^{\text{S/L}}$ at various pressures and temperatures. Zhang et al. (29) reported $D_{\text{mole}}^{\text{S/L}}$ values of 0.54 at (5,000 K, 250 GPa), 0.56 at (6,000 K, 330 GPa), and 0.55 at (6,500 K, 360 GPa), in good agreement with our results at (6,000 K, 323 GPa). Yuan and Steinle-Neumann (27) reported a value of 0.62 at (6,188 K, 330 GPa). In contrast, Liu and Jing (28) reported smaller values, ranging from 0.29 to 0.46 at 6,200 K and 330 GPa. We find significant differences in the underlying ab initio calculations. Both our work and Zhang et al. (29) employed the PAW16 potential for Fe in DFT calculations, which explicitly includes the contributions from inner-shell 3s and 3p electrons. In contrast, the MLPs used in refs. 27 and 28 were trained on DFT data that excluded the 3s electrons. Moreover, ref. 27 employed a fixed electronic temperature of 6,000 K in the DFT calculations, which may affect the temperature dependence of $D_{\text{mole}}^{\text{S/L}}$. It has been shown that inner-shell electrons and the treatment of electronic temperature significantly impact the thermodynamics of Fe (22, 23, 36). Therefore, regardless of the specific thermodynamic methods used to compute $D_{\text{mole}}^{\text{S/L}}$, differences in DFT settings that affect Fe thermodynamics may lead to the variations in the resulting hydrogen partition coefficients.

Given that the phase diagrams at 323 and 360 GPa can be scaled by the melting temperature of Fe, as shown in Fig. 2D, we replot all the partition coefficients as a function of reduced temperature, δT , referenced to the corresponding Fe's melting temperature computed using consistent DFT settings, in Fig. 4B. We find that the simulation data from our work and the ones from refs. 27 and 29 collapse onto a common trend and exhibit good agreement. The results from ref. 28 deviate from other data, likely due to the free-energy calculation methodology used in that study. In experiments, Oka et al. (25) reported a partition coefficient of 0.77 by weight fraction, corresponding to 0.81 by molar ratio, at (1,900 K, 45 GPa). Hikosaka et al. (26) reported a partition coefficient of 0.7 by weight fraction, or 0.77 by mole ratio, at (2,100 K, 56 GPa). Using experimental

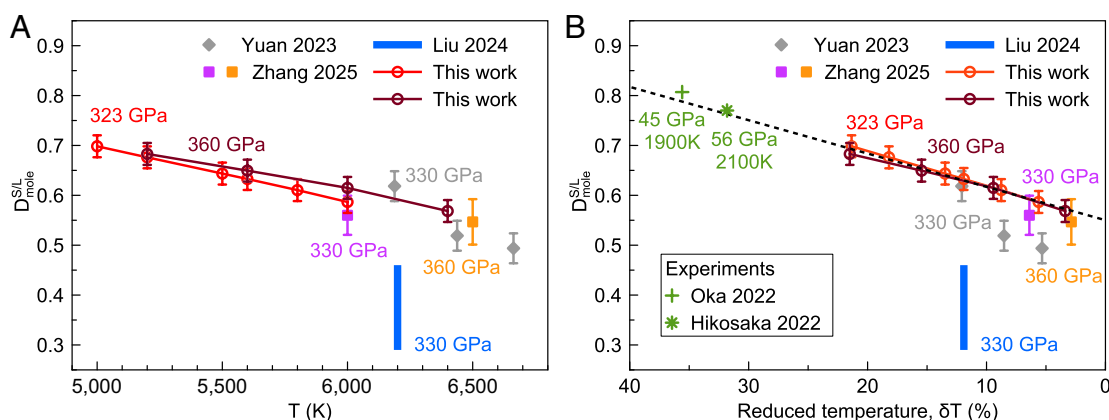


Fig. 4. Hydrogen partition coefficient. (A) Mole partition coefficient between the superionic and liquid phases as a function of temperature. “Yuan 2023,” “Liu 2024,” and “Zhang 2025” refer to ref. 27, ref. 28, and ref. 29, respectively. (B) Mole partition coefficient as a function of reduced temperature, δT . The dashed line represents a linear extrapolation of data from this work at 323 GPa and 360 GPa. “Oka 2022” and “Hikosaka 2022” refer to ref. 25 and ref. 26, respectively. For $T_{\text{m,Fe}}$, we adopt values of 6,357 K at 323 GPa, 6,410 K at 330 GPa, and 6,692 K at 360 GPa for DFT calculations including the inner-shell 3s and 3p electron contribution. For DFT calculations without 3s-electron contribution, a $T_{\text{m,Fe}}$ of 7,038 K at 330 GPa is used as the reference (27).

melting temperatures for Fe ($T_{m,Fe} = 2,950$ K at 45 GPa and 3,080 K at 56 GPa) (37), we find that the experimental data, when plotted as a function of δT , agree remarkably well with the curve extrapolated from our data in Fig. 4B, despite the large differences in pressure and absolute temperature.

The consistency between independent DFT calculations and experimental measurements strongly suggests that reduced temperature, δT , is the key thermodynamic parameter controlling the hydrogen partition coefficient in the Fe–H system. This behavior can be understood by noting that hydrogen is superionic in the solid phase and fully liquid-like in the melt, leading to similar contributions of the hydrogen subsystem to both phases, as manifested in the H–H PPCF shown in Fig. 1A. Thus, the partitioning behavior is primarily governed by the structural and entropic contrast between the solid and liquid Fe sublattices, which is naturally captured by the reduced temperature relative to the melting temperature of Fe.

Because the H partition coefficient depends mainly on reduced temperature and only weakly on pressure (Fig. 4), newly crystallized phase at the ICB is expected to incorporate similar H contents at different stages of IC growth. With continued IC growth, earlier-formed layers become progressively located deeper within the IC. The H contents in these layers should evolve toward the equilibrium radial H gradient (Fig. 3C), driving superionic H outward toward the ICB. Once H reaches the ICB, it preferentially repartitions into the liquid because $D^{S/L} < 1$, further enriching the OC in H. Thus, the present results imply that H exchange between the IC and OC may arise not only from direct partitioning during IC crystallization but also from the continuous redistribution of superionic H within the growing IC. Both processes may contribute to chemical buoyancy, which is believed to be an important energy source driving the geodynamo (38, 39).

Different H contents between the IC and OC can also contribute significantly to the density contrast at the ICB. With increasing H content in the IC, the density contrast becomes larger (SI Appendix, Fig. S19). An H content of 23 atomic percent (at.%), i.e., 0.54 weight percent (wt.%), in the IC is sufficient to reproduce the PREM density contrast by H alone. However, H enrichment also strongly depresses the Fe's melting point. The 23 at.% H content in the IC leads to an ICB temperature of 5,150 K and an outer-core H content of 36 at.% (~1 wt.%). The former is too low compared with the lower bound estimated from CMB temperature constraints (1), while the latter exceeds the upper limit of the H content in the OC based on estimates of H partitioning during core–mantle differentiation (13, 40). Therefore, a more plausible scenario likely requires considering the H phase diagram together with other light elements, especially strongly liquid-preferring elements such as O (24, 41) or C (42–44), to jointly account for the density contrast at ICB.

Seismic observations reveal complex anisotropy in the IC, which is commonly attributed to the lattice-preferred orientation of crystalline phases (6, 45). However, the mechanism responsible for the depth-dependent change in anisotropy, which is weak near the ICB but stronger toward the IC center, remains an open question (46, 47). Recent experiments suggest that the anisotropy of Fe–Si–C alloys can be strongly affected by light-element composition (10). Superionic H can also have a complex effect on the anisotropy of Fe–H alloys, with sensitivity to the chemical composition and the geomagnetic field (48). Our predicted radial gradient of H could naturally contribute to the depth-dependent anisotropy of the IC if lower H concentrations, together with increased pressure, enhance the alloy's elastic anisotropy, an important topic for future work. In addition, the bcc phase can exhibit

elastic anisotropy distinct from that of hcp Fe under IC conditions (49, 50). Although the superionic bcc phase remains metastable in the Fe–H alloy considered here (Fig. 2B), it may be stabilized by additional alloying elements such as Ni (23) and Si (21). Thus, strong hcp–bcc phase competition in multicomponent Fe alloys may lead to coexistence of these phases in the IC, together with a radial gradient of light elements. Such coupled effects can provide a possible mechanism for depth-dependent IC anisotropy, although a quantitative assessment requires further calculations across the relevant pressure, temperature, and composition ranges.

In summary, we computed comprehensive ab initio Gibbs free energies for liquid, superionic hcp, and bcc Fe–H phases and constructed the superionic–liquid phase diagram for pressure–temperature conditions relevant to the IC. Our calculations show that hydrogen can stabilize a superionic bcc phase at sufficiently high temperature and hydrogen content. However, this stability field is superseded by melting, so only the superionic hcp phase coexists with the liquid in the Fe–H system. The superionic–liquid phase diagrams at different inner-core pressures collapse when scaled by the Fe's melting temperature. As a result, the partition coefficient is primarily controlled by the reduced temperature δT and shows minimal sensitivity to pressure under this scaling. Moreover, most previous calculations become mutually consistent, and the extrapolated results agree well with low-pressure experimental data when expressed in terms of this scaling relation.

The phase diagram imposes direct thermodynamic constraints on the equilibrium hydrogen contents of coexisting solid and liquid at the ICB. Through thermochemical equilibrium, it further constrains the hydrogen distribution throughout the IC. These constraints predict a radial hydrogen gradient, with hydrogen enriched toward the ICB relative to the center. Our results, therefore, demonstrate that the nonuniform distribution of superionic hydrogen in the IC is a direct consequence of equilibrium thermodynamics. This mechanism may also apply to the distribution of other light elements, influencing core's composition, dynamics, and evolution.

Materials and Methods

AbInitio Gibbs Free Energy Calculation. The ab initio Gibbs free energies of Fe–H liquid, G^L , at 323 GPa were computed using classical-to-ab initio thermodynamic integration (CATI) calculations (19). The free energies of superionic hcp and bcc Fe–H phases were obtained using a workflow (24) that combines solid–liquid coexistence simulations (51) and CATI, utilizing the thermodynamic relation for the superionic–liquid equilibrium as (24)

$$G^{Sl}(x^{Sl}) = G^L(x^L) - (x^L - x^{Sl}) \frac{\partial G^L(x^L)}{\partial x}, \quad [4]$$

where x^L and x^{Sl} are the equilibrium hydrogen atomic fractions in the liquid and superionic solutions $Fe_{1-x}H_x$, respectively, as determined from superionic–liquid coexistence simulations.

The superionic–liquid coexistence simulations were performed using the LAMMPS software package (52). An embedded-atom method (EAM) potential for the Fe–H system was developed to simulate the superionic and liquid Fe–H phases using an iterative algorithm (53) (see SI Appendix, Note S1 and Fig. S1 for details). The superionic–liquid coexistence model contained 36,960 Fe atoms and up to 11,208 H atoms. Temperature and pressure were controlled using the Nosé–Hoover thermostat and barostat with a damping time of $\tau = 0.1$ ps. The simulations were carried out with a time step of 0.25 fs for 2 ns, sufficient to equilibrate the system and ensure convergence of the superionic–liquid equilibrium.

AIMD simulations were performed using the Vienna ab initio simulation package (VASP) (54). The projected augmented wave (PAW) method (55) was used to describe electron–ion interactions, and the generalized gradient

approximation (GGA) in the Perdew–Burke–Ernzerhof (PBE) (56) form was used for the exchange–correlation functional. We employed the Mermin functional (57, 58) to include the electronic entropy, with the electronic temperature set equal to the ionic temperature in AIMD simulations. Supercells containing 250 to 360 atoms were used to avoid the finite-size effects in the CATI calculation. The time step was set to 0.5 fs for AIMD and 1 fs for CATI simulations. For Fe, the PAW potential with $3d^7 4s^1$ valence electrons was used in AIMD and CATI simulations, while the PAW potential with $3s^2 3p^6 3d^7 4s^1$ valence electrons was used in free energy perturbation (59) (FEP) calculations which is necessary to incorporate the inner-shell electron contributions to the free energy (19, 23). PAW potential with $1s^1$ valence electrons was employed for H. A plane-wave cutoff energy of 400 eV and Γ -point sampling were applied in both AIMD and CATI simulations. A higher ENCUT of 750 eV and a denser k-point mesh of $2 \times 2 \times 2$ were used in FEP calculations. The lattice parameters of liquid, superionic hcp, and superionic bcc phases were adjusted at each temperature and composition to maintain the pressure fluctuations within ± 0.5 GPa over 5 ps of NVT-AIMD simulations. Enthalpy data were collected from AIMD runs lasting more than 8 ps.

Thermodynamic Models. The free energy computed at ($P_0 = 323$ GPa, $T_0 = 6,000$ K) was extended to other (P, T) conditions via a few thermodynamic relations. The free energy was computed along the isothermal line for different pressures via direct integration of equation of states $V(T, P)$ at constant T_0 as

$$G(T_0, P) = G(T_0, P_0) + \int_{P_0}^P V(T_0, P) dP. \quad [5]$$

Here, $V(T, P)$ was computed by AIMD and fitted via the third-order Birch–Murnaghan equation of state (60, 61). More details are in *SI Appendix, Note S3*. The Gibbs–Helmholtz equation was employed to compute the free energy at different T along the isobaric line under constant P_0 as

$$G(T, P_0) = \frac{G(T_0, P_0)}{T_0} T - T \int_{T_0}^T \frac{H(T, P_0)}{T^2} dT, \quad [6]$$

where $H(T, P_0)$ is the enthalpy. The composition-dependent Gibbs free energy for $\text{Fe}_{1-x}\text{H}_x$ is fitted by RK expansion (31) as

$$G(x) = G_{\text{Fe}} + ax + k_B T [(1-x) \ln(1-x) + x \ln x] + x(1-x) \sum_{n=0}^{\infty} (1-2x)^n L_n, \quad [7]$$

where G_{Fe} is the Gibbs free energy of pure Fe. a and L_n are the fitting parameters.

Data, Materials, and Software Availability. The data that support the findings are available on Zenodo with the DOI <https://doi.org/10.5281/zenodo.21222523> (62).

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Supporting Information for Radial gradient of superionic hydrogen in Earth's inner core.

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This PDF file includes:

SI Text Note S1-S4
Figures S1 to S19
SI References

Supporting Information Text

Note S1 | Interatomic potential for Fe-H system

To enable large-scale, long-timescale simulations of superionic–liquid coexistence in the Fe–H system, we developed an interatomic potential with embedded-atom method (EAM) for superionic and liquid Fe–H under Earth’s core conditions. The fitting protocol and analytical form of the potential are described in reference (1). The Fe–Fe interaction was taken from our previous work (2) and kept fixed, while the Fe–H and H–H terms were refitted. Specifically, we fitted the partial pair correlation functions of liquid $\text{Fe}_{0.72}\text{H}_{0.28}$ at 323 GPa and 5500 K, and the composition-dependent volumes for $x_{\text{H}} = 4, 12, 20, 28,$ and 36 at.%. The resulting interatomic potential well reproduces the *ab initio* pair correlation functions and mean-squared displacements for the liquid and superionic hcp/bcc solutions, as shown in Fig. S2.

Note S2 | Free energy calculations of Fe-H binary systems at 323 GPa

Using the interatomic potential, the Gibbs free energy of superionic solutions can be calculated at the classical level using (3)

$$G_{\text{c}}^{\text{SI}}(x_0^{\text{SI}}, T_0) = G_{\text{c}}^{\text{L}}(x_0^{\text{L}}, T_0) - (x_0^{\text{L}} - x_0^{\text{SI}}) \left. \frac{\partial G_{\text{c}}^{\text{L}}(x, T_0)}{\partial x} \right|_{x=x_0^{\text{L}}}, \quad (\text{S1})$$

where x_0^{L} and x_0^{SI} are the equilibrium hydrogen concentrations in the liquid and superionic phases at T_0 , respectively. These compositions were determined from superionic–liquid coexistence (SLC) simulations for both the hcp–liquid and bcc–liquid systems. The coexistence cell contained 36,960 Fe atoms and up to 11,208 H atoms; the H content was adjusted to achieve coexistence at each temperature. Initially, H atoms are distributed as uniformly as possible between the solid and liquid regions. Each simulation was run for 2 ns, during which H atoms diffused throughout the cell, providing sufficient statistics to extract the equilibrium compositions of the superionic (hcp/bcc) and liquid phases. Representative coexistence configurations and the corresponding H concentration profiles at 5500 K are shown in Fig. S3A–D, and the resulting EAM phase diagrams are shown in Fig. S3E and S2F. We further tested a configuration in which all H atoms are initially placed in the liquid phase. The results are consistent with those from the uniform distribution, as shown in Fig. S4. Finite-size tests show that systems with more than 40,000 atoms are sufficient to converge the solidus and liquidus within 0.0003, as shown in Fig. S5.

Next, we computed $G_{\text{c}}^{\text{L}}(x, T)$ for solving Eq. (S1) using a non-equilibrium thermodynamic integration method with the Uhlenbeck-Ford model (UFM) as the reference state (4–6). Calculations were performed for a set of selected compositions and temperatures, and the resulting liquid Gibbs free energies are shown in Fig. S6A. The Gibbs free energies of the liquid were fitted using Redlich-Kister (RK) expansion (Eq. (7) in main text) with two RK coefficients, with a fitting error of ~ 0.4 meV/atom. We then expanded $G_{\text{c}}^{\text{SI}}(x_0^{\text{SI}}, T_0)$ to other temperature range using the Gibbs-Helmholtz equation (Eq. (6)). The resulting Gibbs free energies for superionic hcp and bcc phases are shown in Fig. S6B and S6C, respectively. The $G_{\text{c}}^{\text{SI}}(x, T_0)$ can also be fitted by RK expansion with two RK coefficients, with a fitting error of 0.9 meV/atom. Using $G_{\text{c}}^{\text{L}}(x, T_0)$ and $G_{\text{c}}^{\text{SI}}(x, T_0)$, we determined the equilibrium compositions x_0^{SI} and x_0^{L} at temperature T_0 via the common-tangent construction, as shown in Fig. S6D. The solidus and liquidus obtained from Gibbs free energy calculations are consistent with those from the SLC simulations, as shown in Fig. S3E. The same procedure was applied to the bcc–liquid coexistence, yielding the phase boundary shown in Fig. S3F.

With the classical Gibbs free energies of liquid, superionic hcp and bcc phases, we performed classical-to-*ab initio* thermodynamic integration (CATI)(3, 7), using the Fe-H EAM potential as a reference state. The *ab initio* Gibbs free energy, G_{A} , can be computed by

$$G_{\text{A}} = G_{\text{c}} + f_{\text{PV}} + f_{\text{TI}}, \quad (\text{S2})$$

where f_{TI} is the Helmholtz free energy difference and f_{PV} is the PV contribution. The final TI results are independent of how accurately the reference potential reproduces energies and forces, as long as the integration path is sufficiently smooth and does not cross a phase transition(3, 8). Figure S8 summarizes the CATI results for the liquid and superionic hcp/bcc phases at 5500 K over a range of H compositions. The integration volumes were taken from AIMD at 323 GPa (Fig. S7A, S7D, and S7G). The thermodynamic-integration path from the classical to the *ab initio* Hamiltonian is smooth (Fig. S8A, S8C, and S8E), which reduces numerical uncertainty and allows an accurate evaluation of the free-energy differences(7). We further applied free energy perturbation (FEP) to account for the 3s3p semicore contributions, which are required for accurate free energy calculations under inner-core conditions(2, 7, 9) (see in Fig. S7C, S7F and S7I). The temperature range was extended to 5000–6000 K using Eq. (6), with the corresponding

enthalpies shown in Fig. S7B, S7E, and S7H. Finally, the *ab initio* solidus and liquidus were obtained from the common-tangent construction between $G_{\mathcal{A}}^L(x)$ and $G_{\mathcal{A}}^{SI}(x)$, as shown in Fig. S9.

Note S3 | The Equation of State over the pressure–temperature range of 323–360 GPa and 5000–6800 K

Based on Eq. (5) in main text, once the Gibbs free energy is known at a given pressure and temperature, the equation of state $V(T, P)$ enables extrapolation to other pressures. Directly performing AIMD simulations over a wide (T, P) space to obtain volumes, $V(T, P)$, is prohibitively expensive; therefore, we employed a volume interpolation model.

Taking superionic hcp $\text{Fe}_{288}\text{H}_{72}$ phase as an example, we first performed AIMD simulations to obtain volumes at 323 GPa over 5000–6000 K and at 360 GPa over 5200–6800 K (Fig. S10A). At each pressure, we fitted $V(T)$ with a linear function to extract the thermal expansion coefficient, achieving a fitting error of $\sim 0.002 \text{ \AA}^3/\text{atom}$ ($\sim 0.04\%$). Assuming a linear pressure dependence of the expansion coefficient, we interpolated between the values at 323 and 360 GPa to obtain coefficients at intermediate pressures. We further computed three additional volumes at 6000 K and fitted the 6000 K isotherm using a third-order Birch–Murnaghan equation of state (EoS)(10, 11) to the $V(T = 6000 \text{ K}, P)$ data (Fig. S10B). Combining the EoS with the interpolated expansion coefficients yields $V(T, P)$ at intermediate pressures over 5000–6800 K as

$$V(T, P) = V(T_0, P) + \left[\left(\frac{\alpha_2 - \alpha_1}{360 - 323} \right) P + \left(\frac{360\alpha_1 - 323\alpha_2}{360 - 323} \right) \right] (T - T_0), \quad (\text{S3})$$

where $V(T_0, P)$ is the volume at $T_0 = 6000 \text{ K}$ from third-order Birch–Murnaghan fit; α_1 and α_2 are the expansion coefficient at 323 and 360 GPa, respectively. Similar calculations were performed for the liquid phase with compositions of $x=0$ –36 at.%, the superionic hcp phase with $x=0$ –20 at.%, and the superionic bcc phase with $x=0$ –10 at.%, as shown in Fig. S11–S13.

To assess the accuracy of the volumes obtained using the above approach, we performed independent calculations for liquid $\text{Fe}_{0.64}\text{H}_{0.36}$ and superionic hcp $\text{Fe}_{0.8}\text{H}_{0.2}$ at 340 GPa and temperatures of 5200 and 6400 K. The predicted and directly calculated volumes agree within $\sim 0.002 \text{ \AA}^3/\text{atom}$ (Fig. S14).

Note S4 | Uncertainties in the phase diagram

The uncertainties in the phase diagram (Fig. 1D in maintex) and the partition coefficient (Fig. 4 in maintex) originate from the errors in the free energies. According to Eq. S2, the uncertainty in the *ab initio* free energy mainly has two contributions. The first is the uncertainty in the classical free energy G_C , which, for the superionic phase, arises from the uncertainties in $G_C^L(x^L)$ and in the composition difference $x^L - x^{SI}$ in Eq. S1. The second contribution comes from the uncertainties in the correction terms in f_{PV} and f_{TI} .

To evaluate the uncertainty in $G_C^L(x^L)$, denoted as $\delta_{G_C^L}$, we performed three independent calculations using a 32,000-atom system, obtaining an uncertainty of 0.04 meV/atom. We also conducted three independent superionic-liquid coexistence simulations using a 43,448-atom system (Fig. S5), which yielded an uncertainty of 0.0005 in the composition difference $x^L - x^{SI}$, denoted as δ_x . This gives an uncertainty of

$$\delta_{G_C^{SI}} = 1.7 \text{ meV/atom, calculated as } \delta_{G_C^{SI}} = \sqrt{\delta_x^2 \left(\frac{\partial G_C^L(x)}{\partial x} \Big|_{x=x_{C0}^L} \right)^2 + \delta_{G_C^L}^2}, \text{ where } \frac{\partial G_C^L(x)}{\partial x} \Big|_{x=x_{C0}^L} = -3.0134 \text{ eV/atom.}$$

The uncertainties in f_{PV} and f_{TI} were estimated using the method used in our previous work, giving errors of 0.2 and 0.6 meV/atom, respectively. Therefore, the uncertainties in $G_{\mathcal{A}}^L$ and $G_{\mathcal{A}}^{SI}$ are obtained from

$$\delta_{G_{\mathcal{A}}} = \sqrt{(\delta_{G_C})^2 + (\delta_{PV})^2 + (\delta_{TI})^2}, \text{ yielding 0.6 and 1.8 meV/atom, respectively. Uncertainties of 1.8 and 0.6 meV/atom in the free energies of the superionic and liquid phases propagate to uncertainties of 0.005 and 0.006 in the solidus and liquidus, respectively (Fig. 1). Based on the error propagation formula } \sigma_{D^S/I} =$$

$$D^{S/I} \sqrt{\left(\frac{\sigma_{x^{SI}}}{x^{SI}} \right)^2 + \left(\frac{\sigma_{x^L}}{x^L} \right)^2}, \text{ the uncertainty in the partition coefficient in Fig. 4 is estimated to be 0.026.}$$

Figures

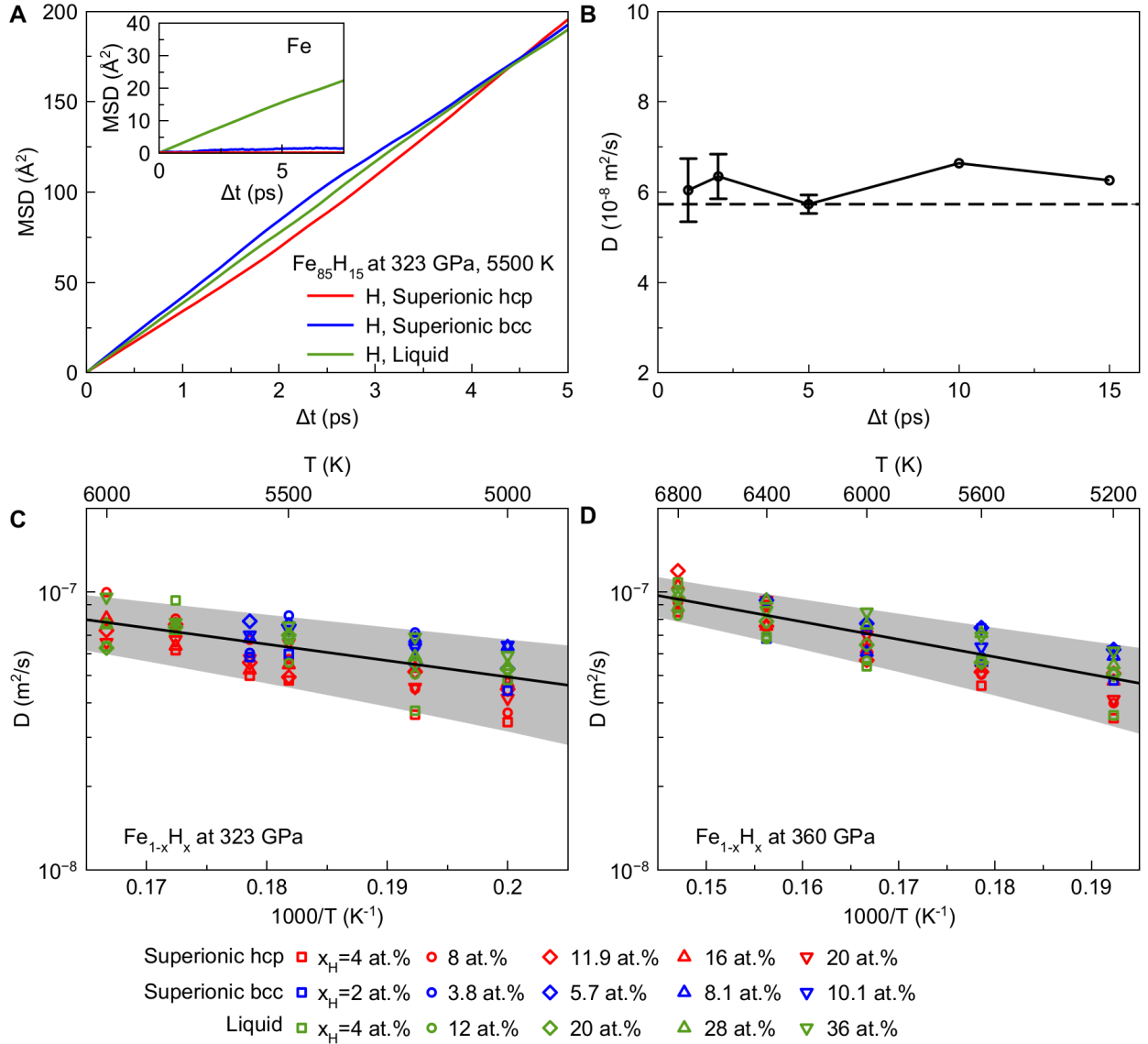


Fig. S1. Diffusive behavior of hydrogen in superionic Fe–H phases. **A.** Mean square displacement (MSD) of Fe (inset) and H atoms in superionic hcp, superionic bcc, and liquid phases at 323 GPa and 5500 K, with a hydrogen concentration of 15 at.%. **B.** Time-convergence tests for the diffusion coefficient calculations. Error bars at 1 ps, 2 ps, and 5 ps are obtained from three independent AIMD simulations. **C.** and **D.** Hydrogen diffusion coefficients for superionic hcp, superionic bcc, and liquid phases with varying compositions as a function of temperature at 323 GPa and 360 GPa, respectively. The line shows an Arrhenius fit, with the shaded band indicating variations due to different chemical compositions.

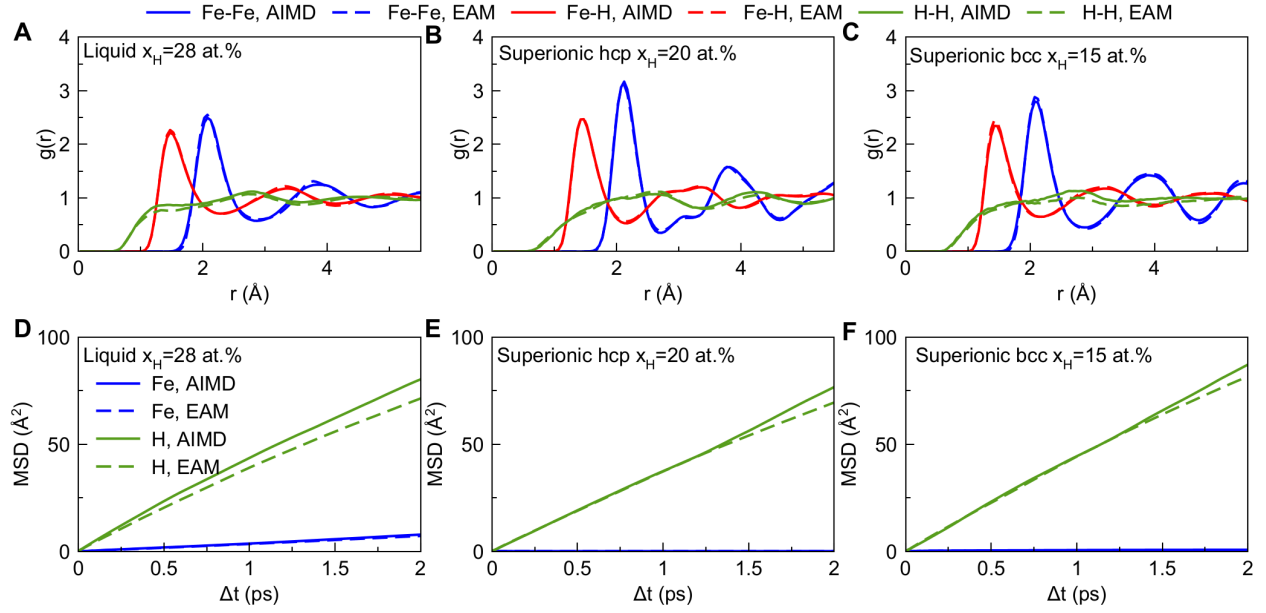


Fig. S2. A–C. Partial pair correlation functions (PPCF) for liquid, superionic hcp, and superionic bcc Fe–H, respectively. **D–F.** Mean-squared displacements (MSDs) for liquid, *superionic* hcp, and superionic bcc Fe–H, respectively. All results are shown at 323 GPa and 5500 K. Solid and dashed lines correspond to AIMD and EAM-MD results, respectively.

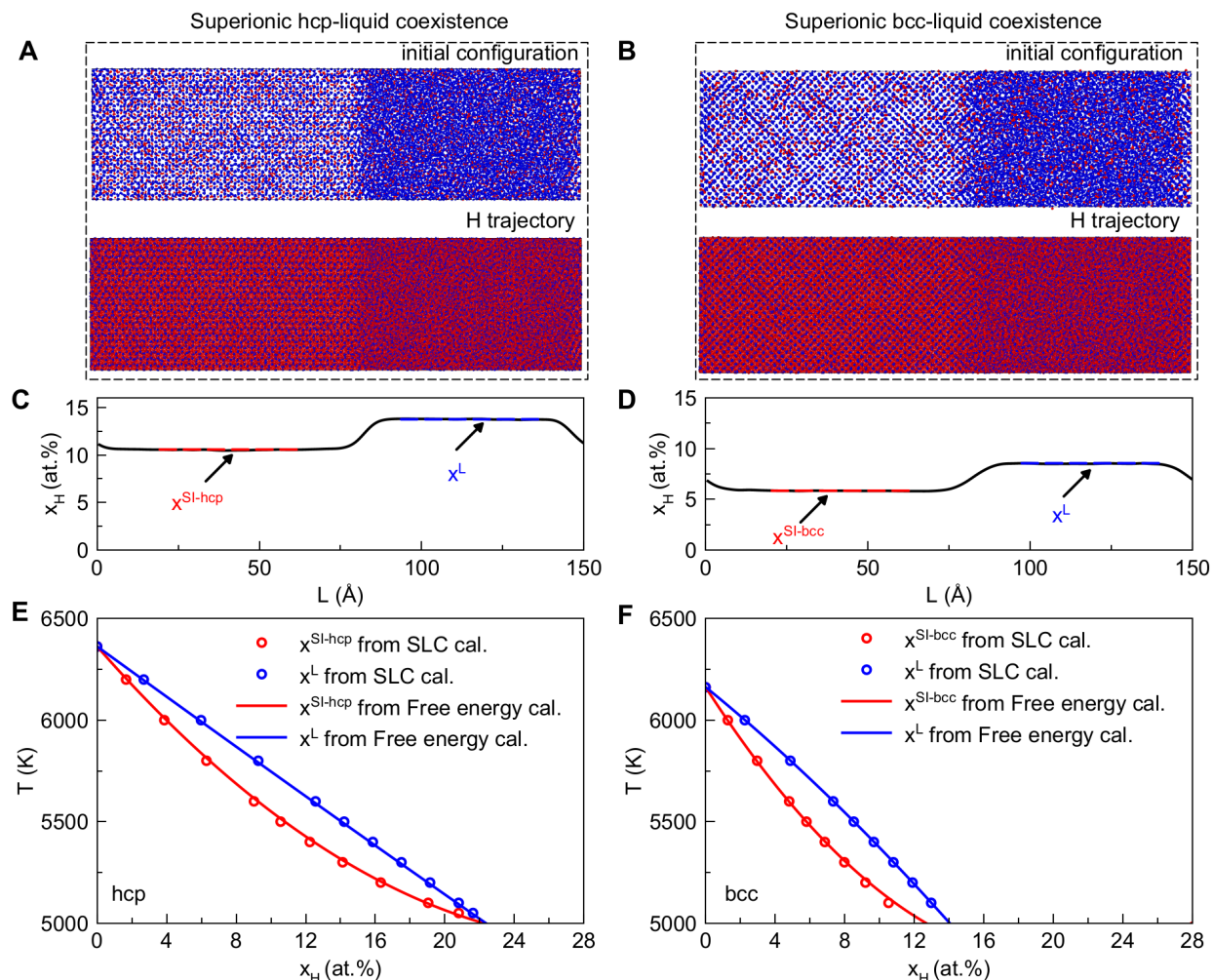


Fig. S3. Classical Fe-H phase diagrams from superionic-liquid *coexistence* simulations at 323 GPa. **A.** hcp-liquid two-phase simulation. Top: initial configuration, where blue and red spheres represent Fe and H atoms, respectively. Bottom: H-atom trajectories over 2 ns (red lines), with Fe atoms shown as blue spheres. **B.** bcc-liquid two-phase simulation. Top: initial configuration (Fe in blue; H in red). Bottom: H-atom trajectories over 2 ns (red lines), with Fe atoms shown as blue spheres. **C** and **D.** Hydrogen spatial distributions during the last 1 ns of the simulations for the hcp-liquid and bcc-liquid coexistence runs, respectively. **E** and **F.** Classical phase diagrams at 323 GPa for the superionic hcp-liquid and superionic bcc-liquid systems, respectively. Symbols denote results from SLC simulations, and solid curves denote Gibbs free-energy calculations.

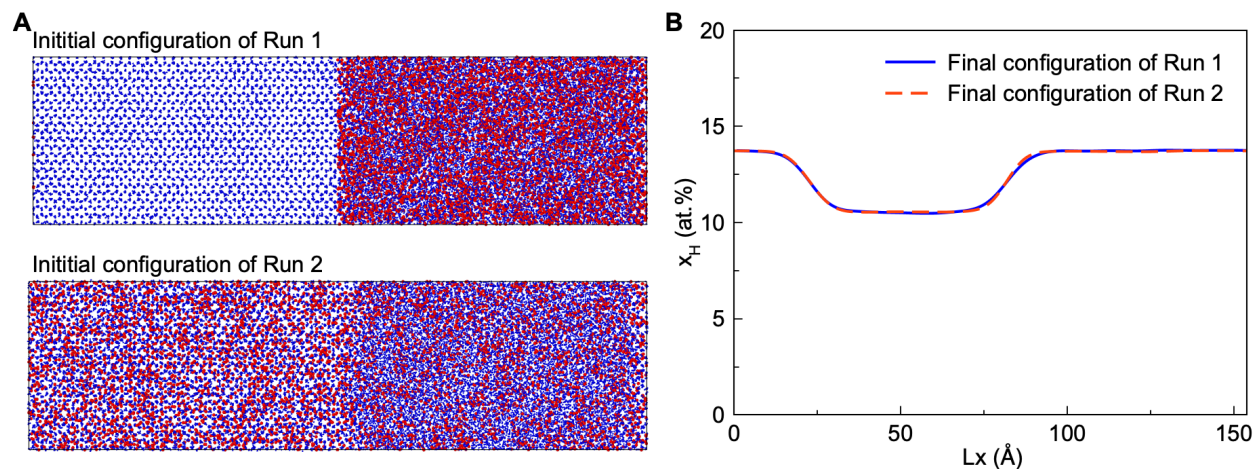


Fig. S4. Dependence of the two-phase simulations on the initial hydrogen distribution. **A.** Two MD runs with different initial configurations: (top) all H atoms initially placed in the *liquid* region and (bottom) H atoms uniformly distributed across both phases. Blue and red spheres denote Fe and H atoms, respectively; atomic radii are rescaled for clarity. **B.** Equilibrium H concentration profiles along the x-direction after 2 ns of simulation. Both initial conditions converge to the same final distribution, demonstrating that the results are independent of the initial hydrogen configuration.

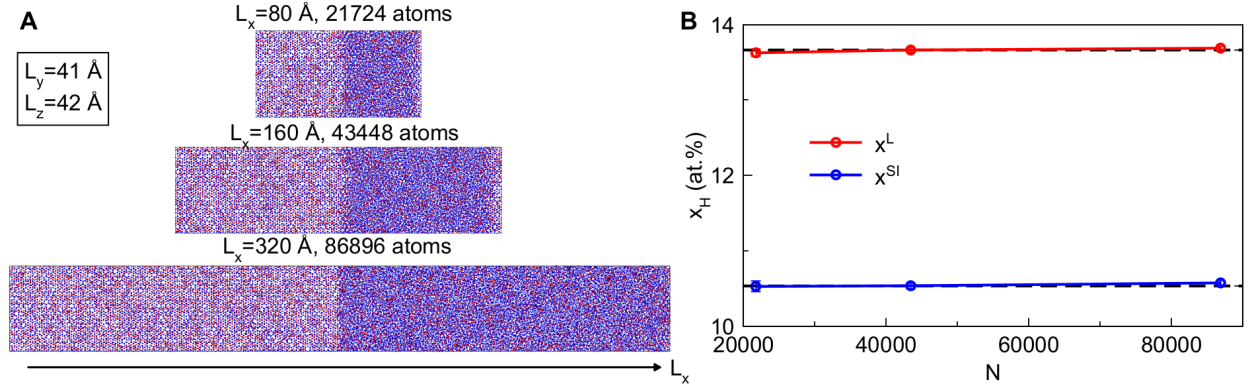


Fig. S5. Size effect on the superionic–liquid coexistence simulation. **A.** Initial simulation configurations with different cell lengths perpendicular to the interface. Blue and red spheres denote iron and hydrogen atoms, respectively; atomic radii are rescaled to better visualize the initial hydrogen distribution. **B.** Solidus (x^{SI}) and liquidus (x^L) at 5500 K obtained using different simulation sizes. Error bars denote the standard deviation from three independent runs and are smaller than the symbol size. The dashed line indicates the system size used in the thermodynamic calculations presented in the main text. The results show that both x^{SI} and x^L are converged with respect to system size.

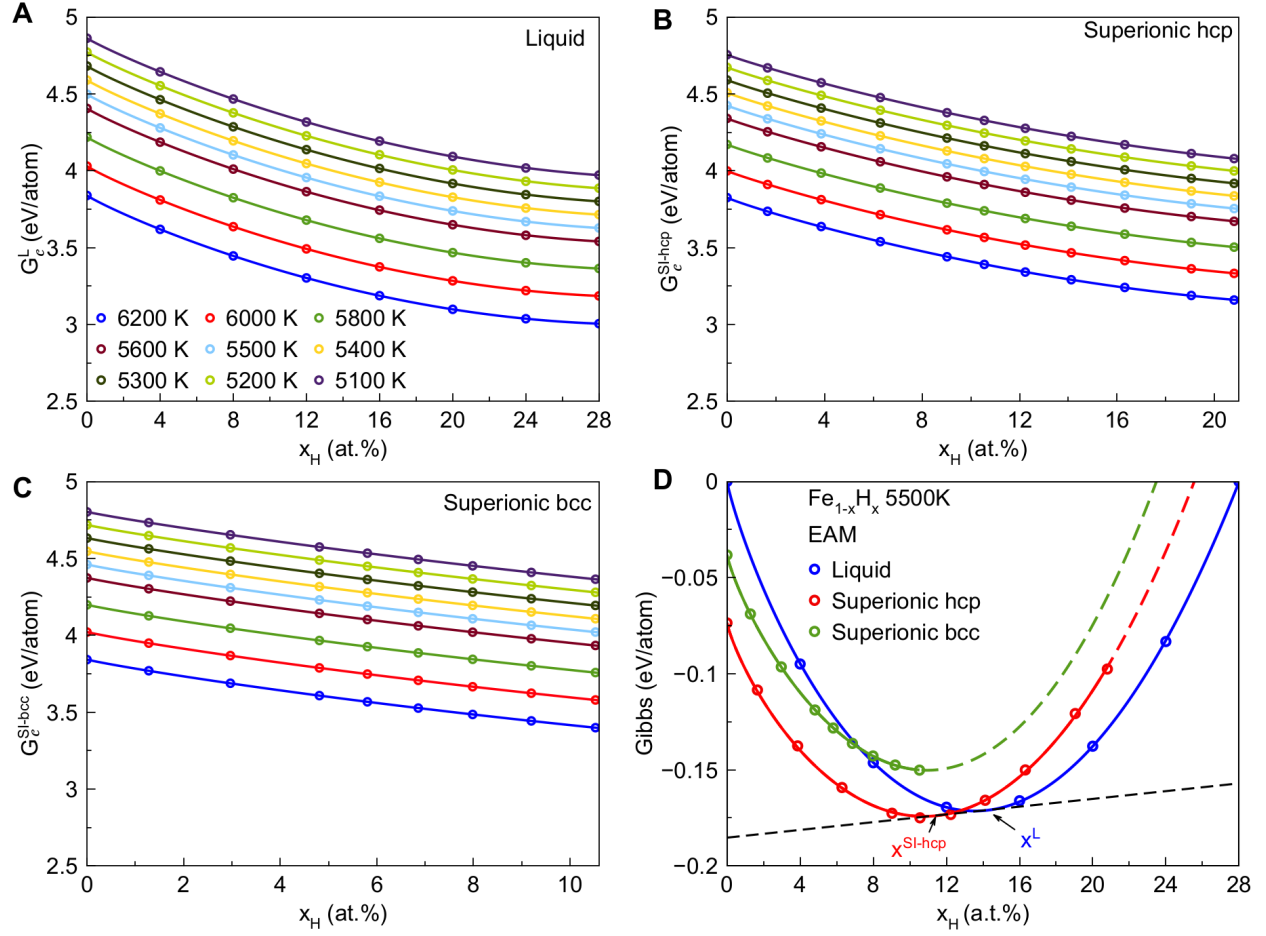


Fig. S6. A–C. Classical Gibbs free energies of liquid, superionic hcp, and superionic bcc Fe_{1-x}H_x solutions at 323 GPa, respectively. Solid curves show fits to RK expansion (12). **D.** Gibbs free energies of three phases at 5500 K. The black dashed line marks the common cotangent between liquid and superionic hcp curves, from which the solidus and liquidus are determined.

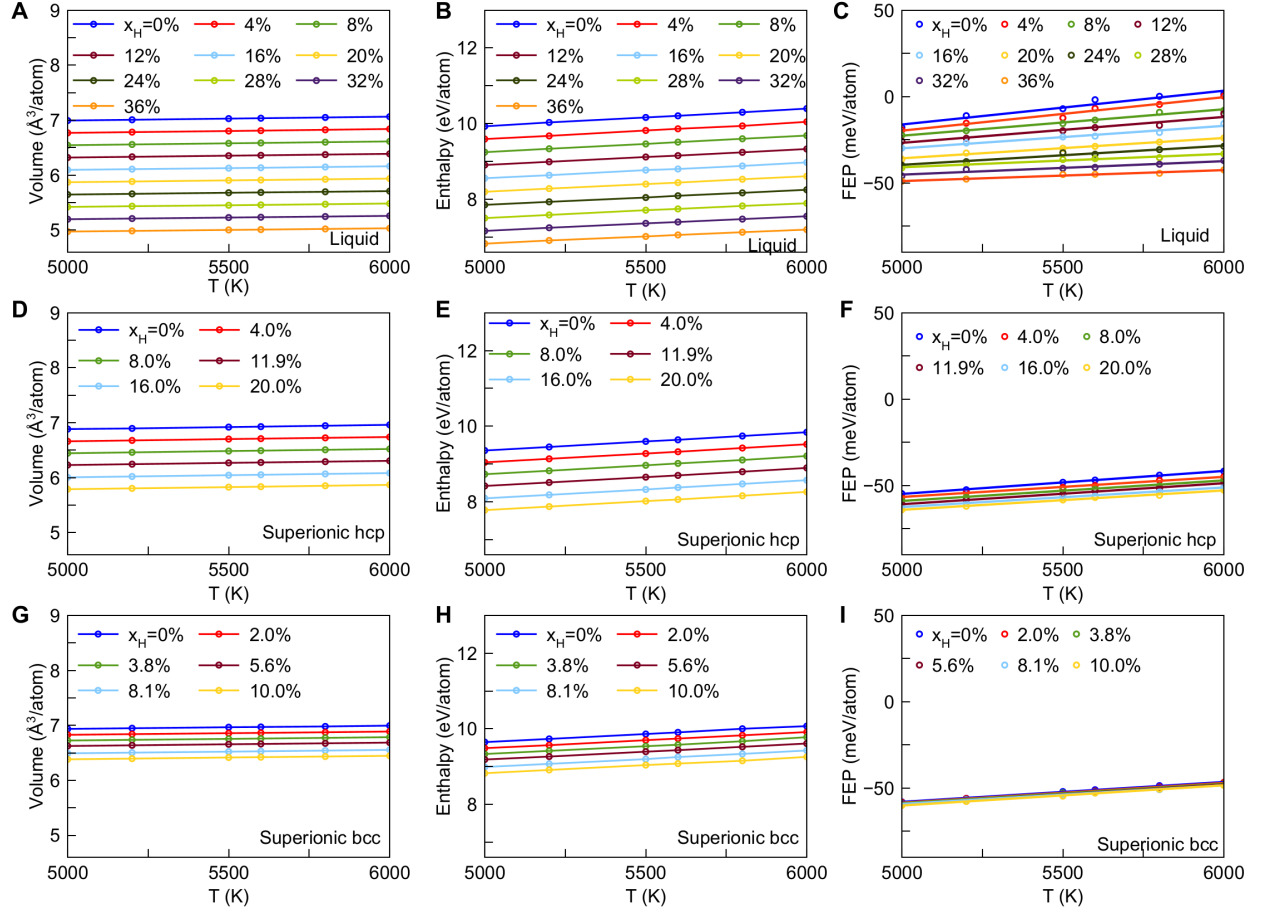


Fig. S7. A–C. Volumes, enthalpies, and FEP contributions for liquid $\text{Fe}_{1-x}\text{H}_x$ solutions. **D–F.** Superionic hcp $\text{Fe}_{1-x}\text{H}_x$ solutions. **G–I.** Superionic bcc $\text{Fe}_{1-x}\text{H}_x$ solutions. All results are from AIMD simulations at 323 GPa.

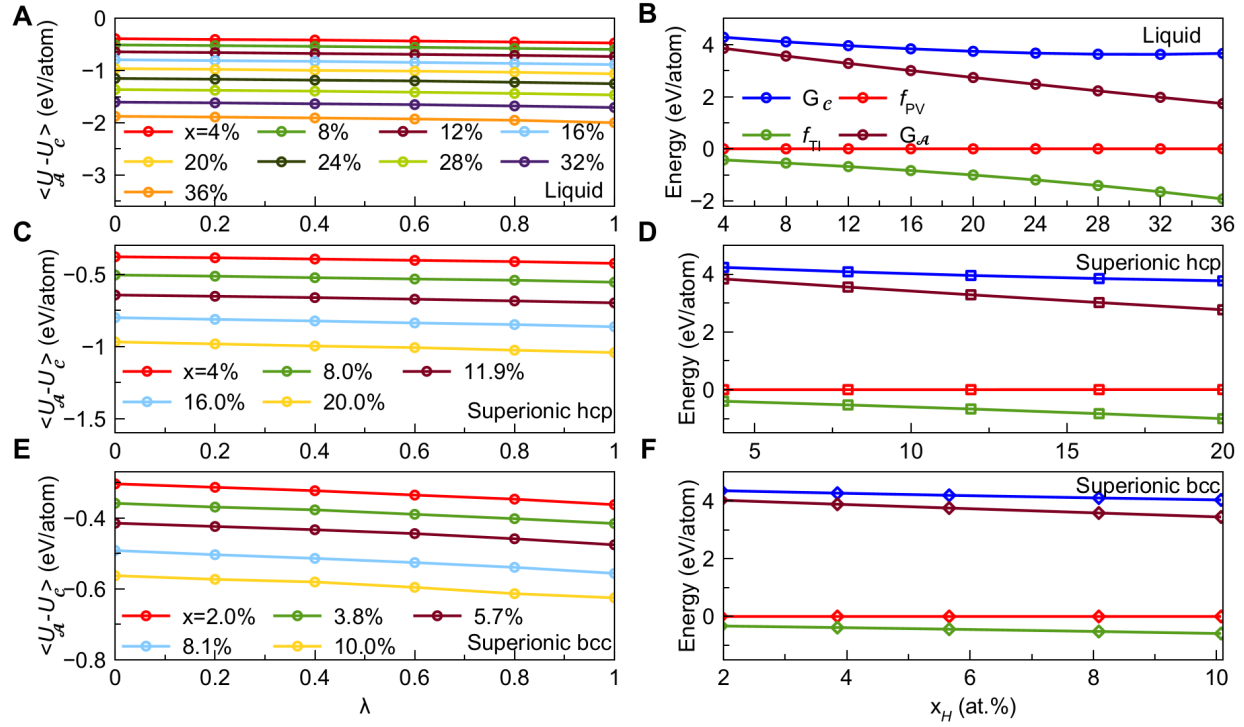


Fig. S8. Classical-to-*ab initio* thermodynamic integration (CATI) for **A.** liquid, **C.** superionic hcp, and **E.** superionic bcc $\text{Fe}_{1-x}\text{H}_x$ solutions at 5500 K. **B.** **D.** and **F.** show the corresponding energy terms in Eq. (S2) for the liquid, superionic hcp, and superionic bcc $\text{Fe}_{1-x}\text{H}_x$ solutions, respectively, at 5500 K.

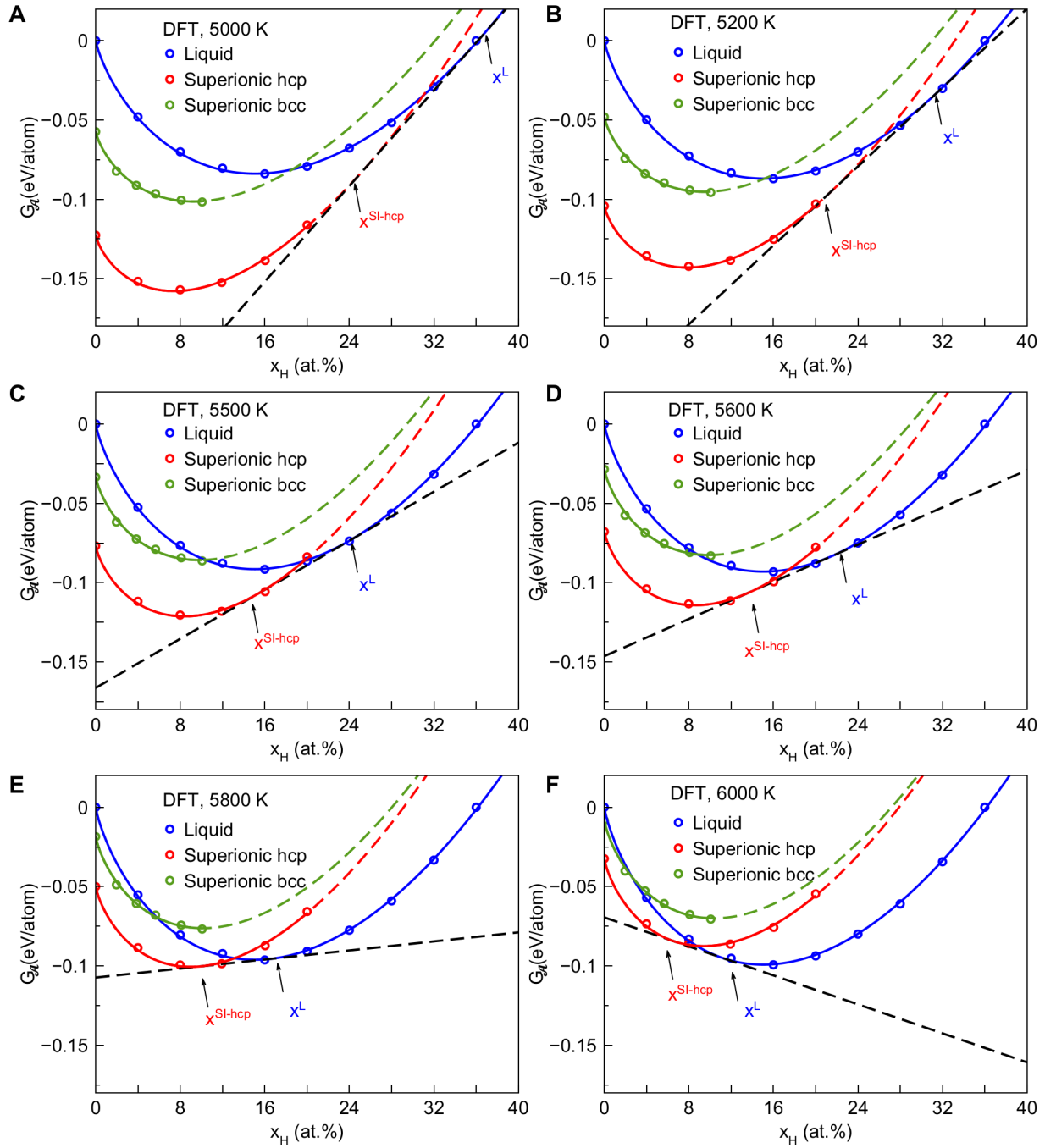


Fig. S9. *Ab initio* Gibbs free energy of $\text{Fe}_{1-x}\text{H}_x$ solutions at 323 GPa 5000-6000 K. Solid curves are Redlich-Kister fits (12) using two fitting parameters, with a fitting error of ~ 0.9 meV/atom; dashed curves show extrapolations. Black dashed lines mark the common tangent constructions used to determine the solidus and liquidus.

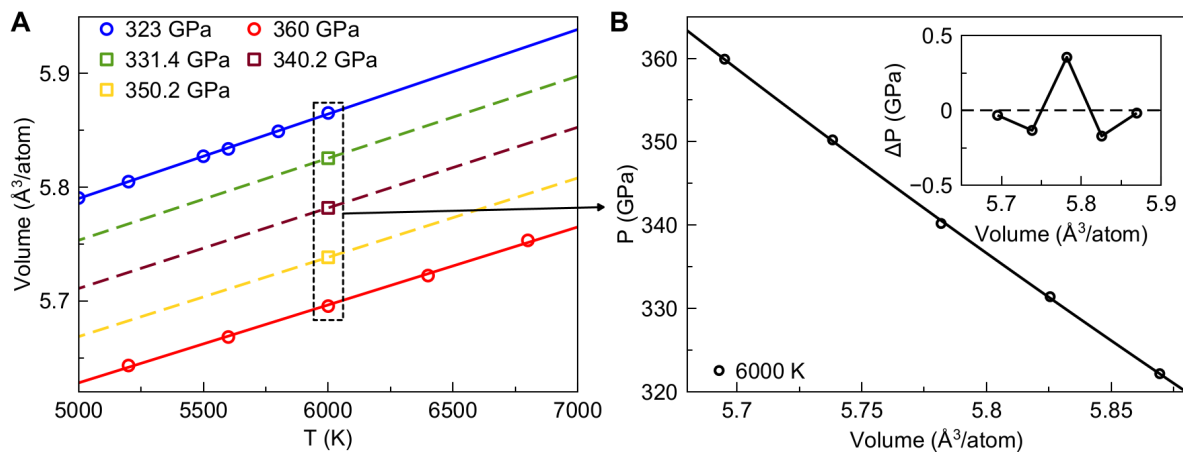


Fig. S10. A. Volumes of superionic hcp $\text{Fe}_{288}\text{H}_{72}$ ($\text{Fe}_{0.8}\text{H}_{0.2}$) phase at multiple temperatures and pressures. Circular and square are volumes directly obtained from AIMD. Solid blue and red lines indicate fits of volume versus temperature at 323 GPa and 360 GPa, respectively. Dashed lines show the predicted volume–temperature relations at intermediate pressures, using thermal expansivities obtained by linear interpolation (in pressure) between the 323 GPa and 360 GPa slopes. **B.** At 6000 K, volume as a function of pressure is interpolated with a third-order Birch–Murnaghan equation of state (10, 11). Inset: deviation between the fit and the AIMD results.

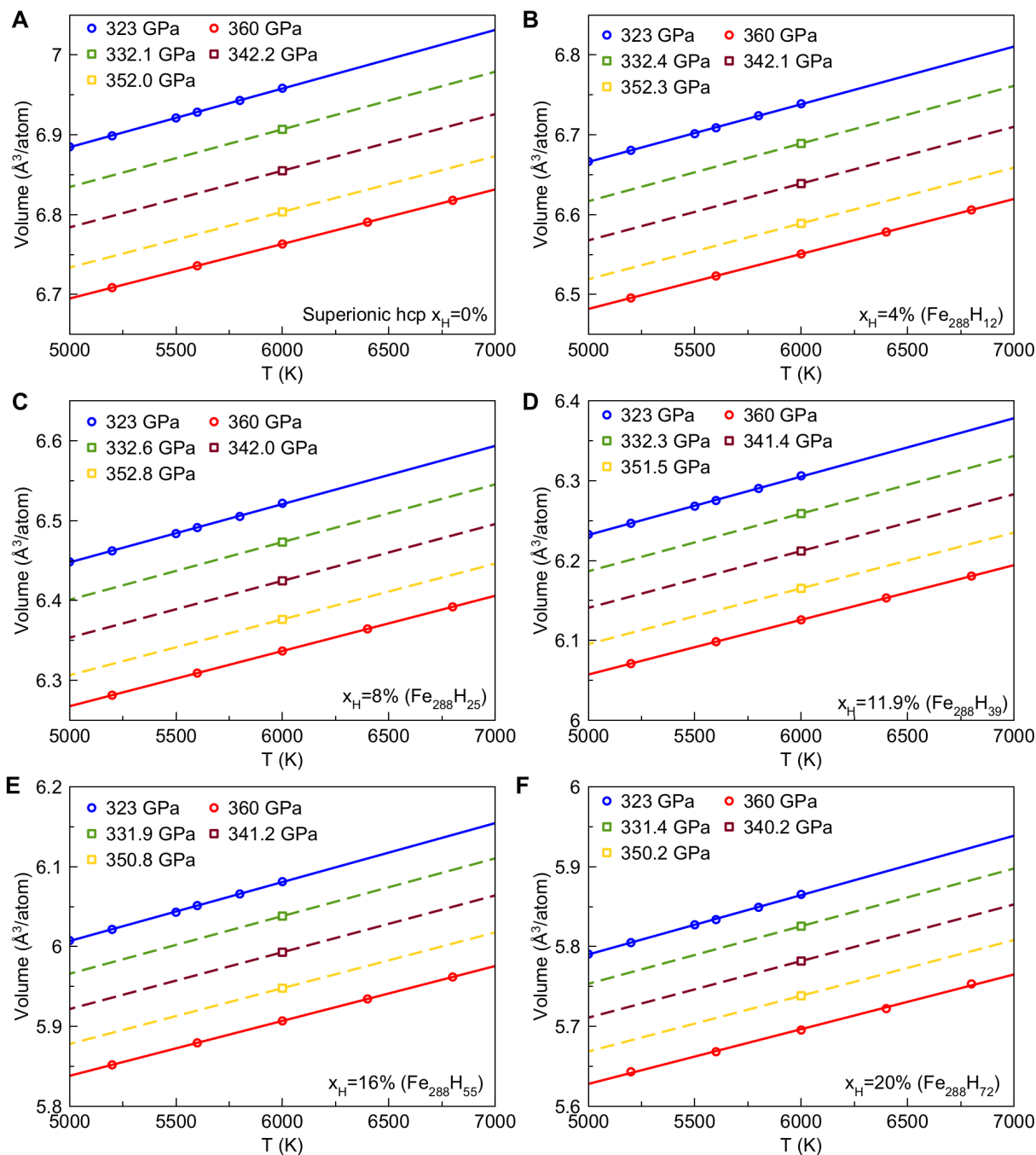


Fig. S11. Volumes of superionic hcp Fe–H solutions with different H contents over 323–360 GPa and 5000–6800 K; interpolation details are described in Note S3.

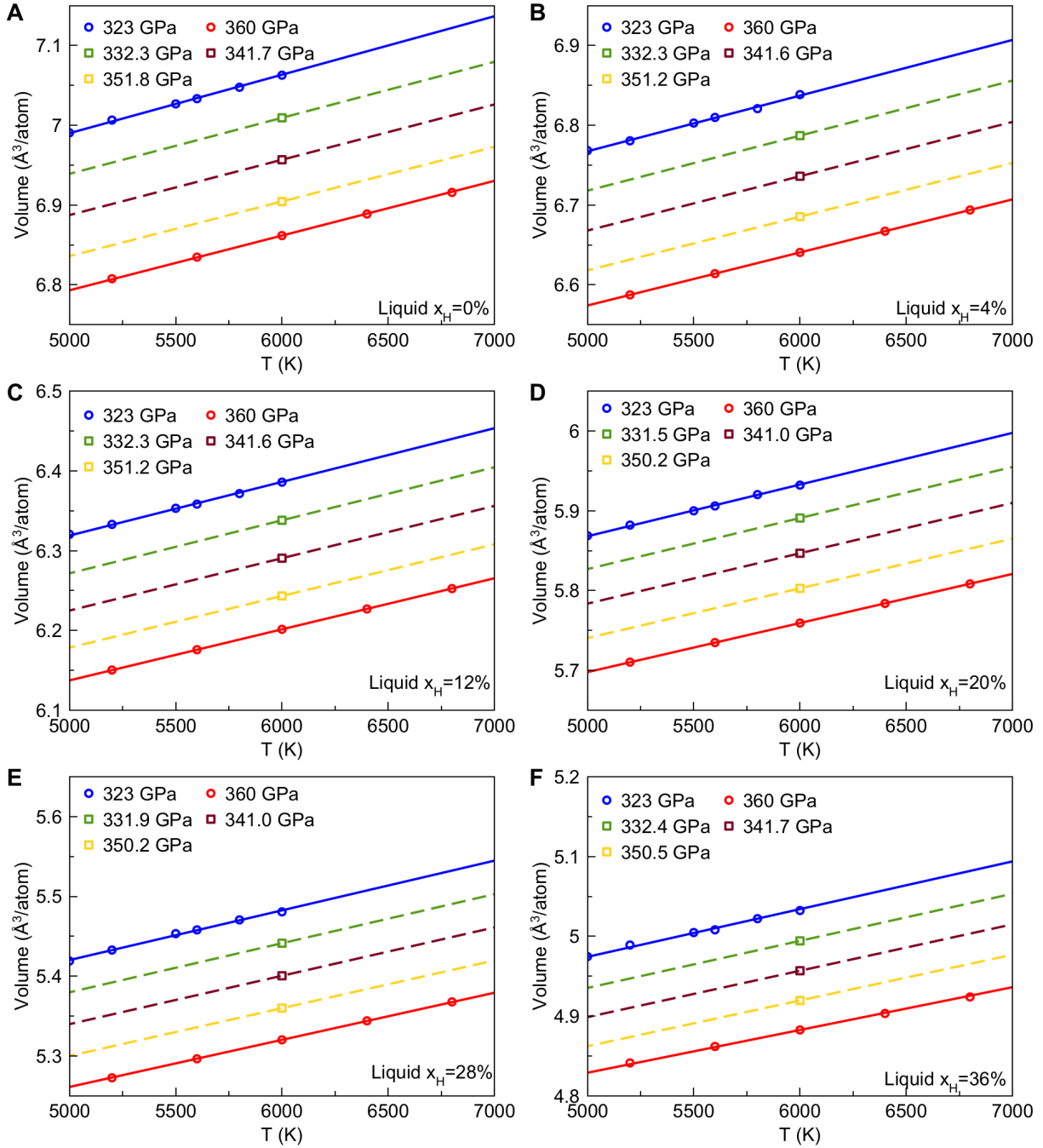


Fig. S12. Volumes of liquid Fe–H solutions with different H contents over 323–360 GPa and 5000–6800 K; interpolation details are described in Note S3.

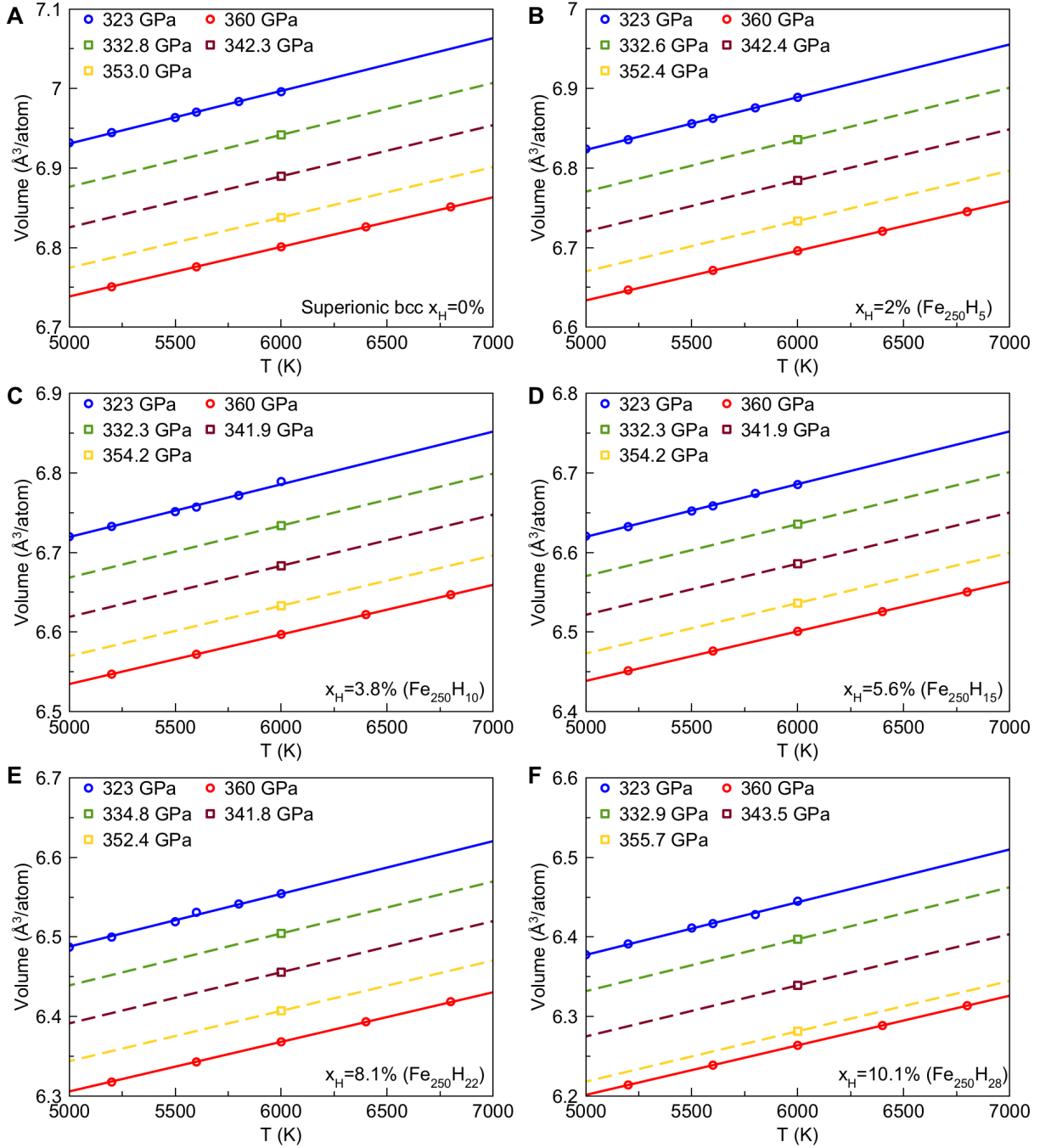


Fig. S13. Volumes of superionic bcc Fe-H solutions with different H contents over 323–360 GPa and 5000–6800 K; interpolation details are described in Note S3.

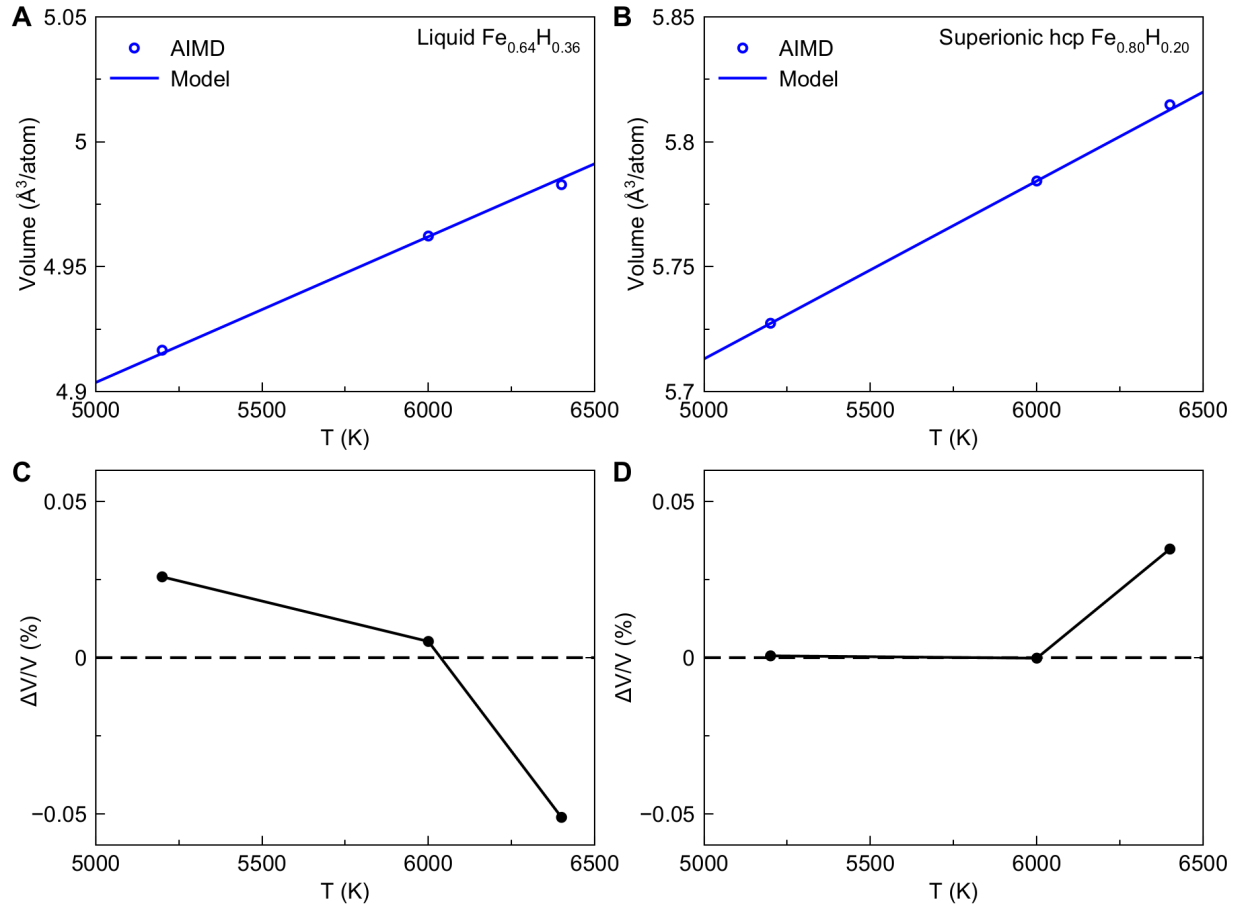


Fig. S14. Difference between the volumes predicted by the interpolation model (Note S3) and those obtained from AIMD simulations at 340 GPa.

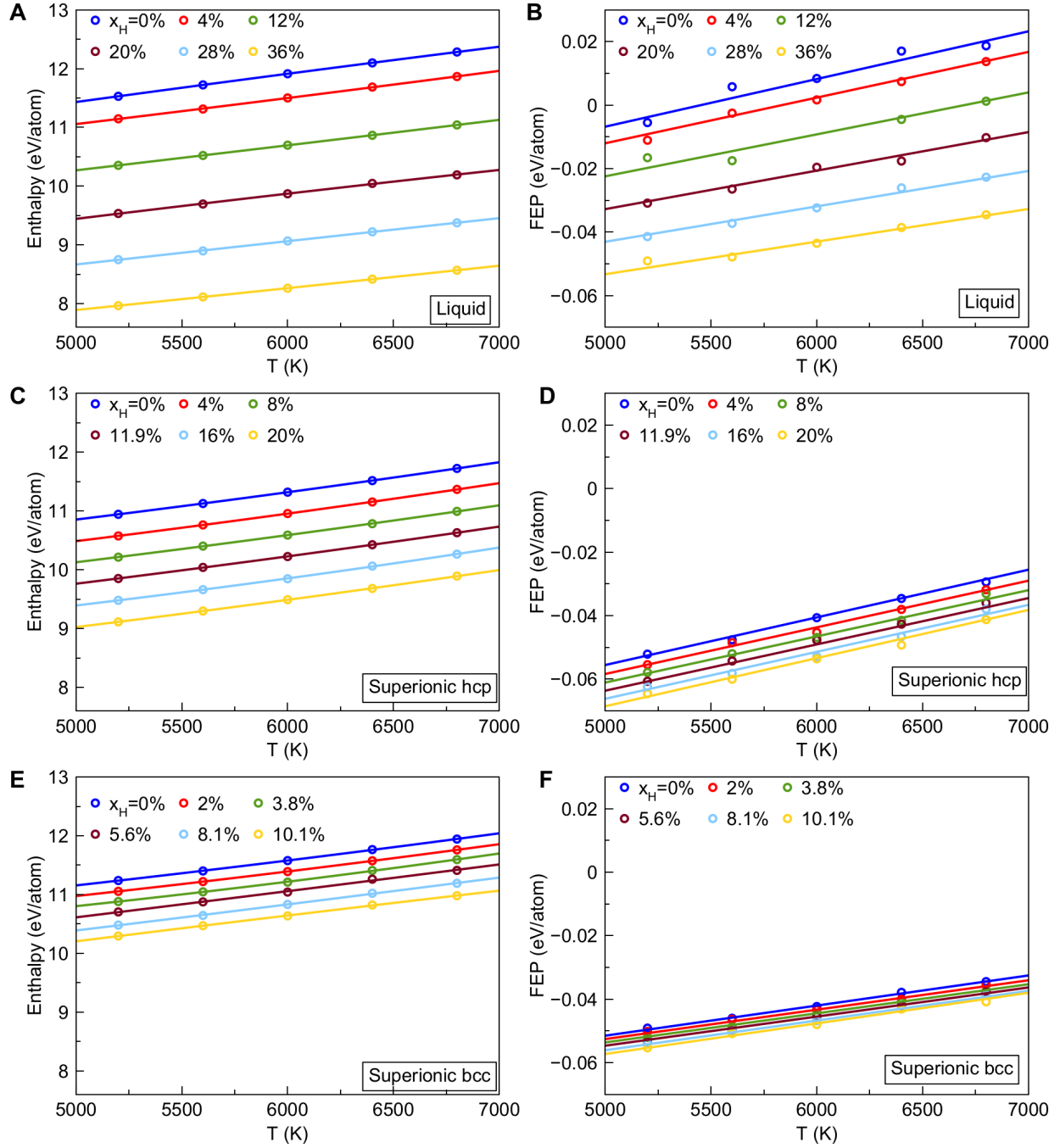


Fig. S15. Enthalpies and FEP contributions for **A–B.** liquid, **C–D.** superionic hcp, and **E–F.** superionic bcc $\text{Fe}_{1-x}\text{H}_x$ solutions from AIMD simulations at 360 GPa.

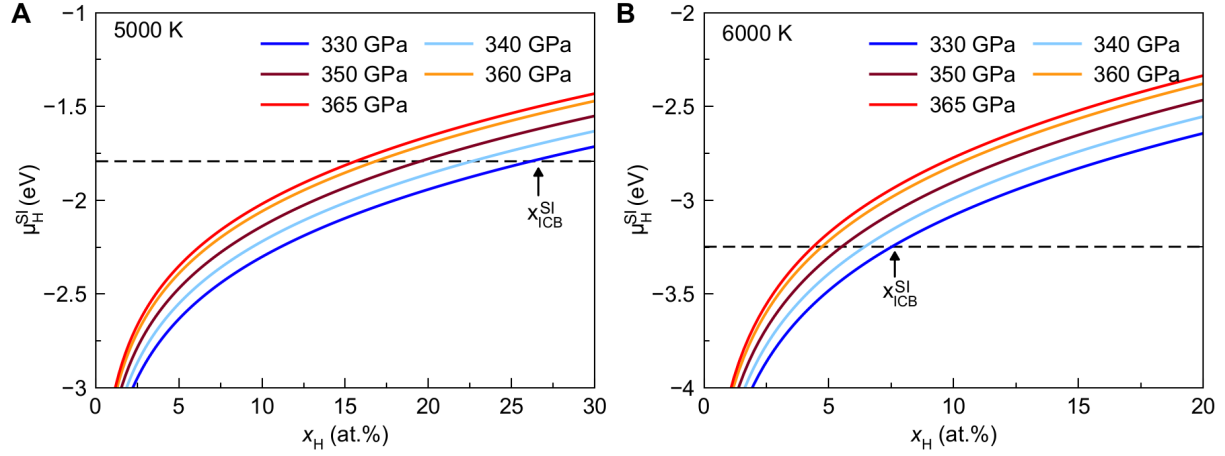


Fig. S16. Hydrogen chemical potential in the superionic hcp phase calculated from Eq. (2) at **A.** $T_{ICB} = 5000$ K, and **B.** $T_{ICB} = 6000$ K. The black arrow marks the hydrogen content of the solid at the ICB (330 GPa), determined from the phase diagram in Fig. 2C ; the dashed line indicates the corresponding chemical potential, $\mu_H^{SI}(x_{ICB}^{SI}, T_{ICB}, P_{ICB})$.

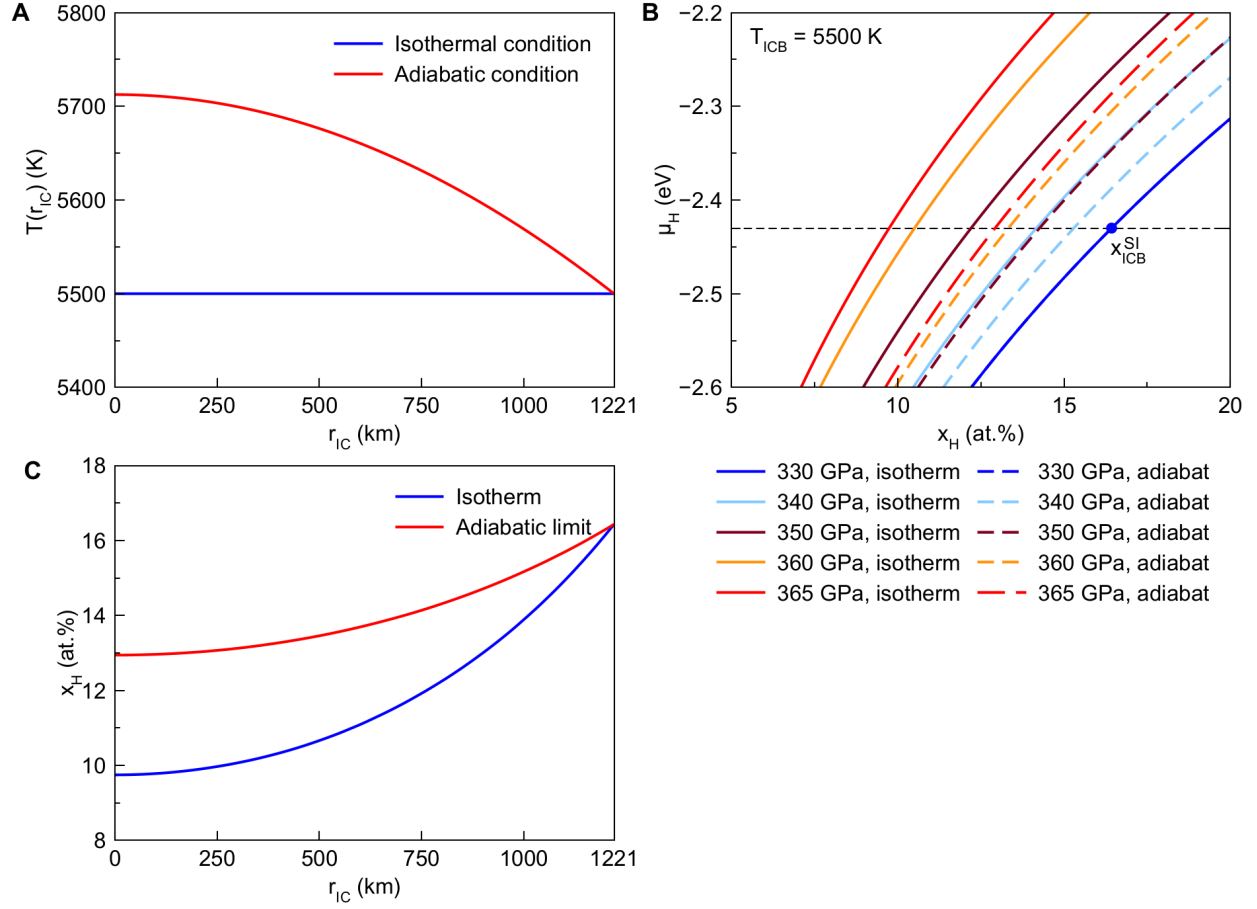


Fig. S17. Effect of inner-core temperature profile on *hydrogen* distribution. **A.** Radial temperature profiles under isothermal and adiabatic assumptions. The adiabatic profile follows $T(r) = T_{ICB} \exp\left(\gamma \frac{r_{ICB}^2 - r^2}{L_p^2}\right)$, where $\gamma=1.5$ and $L_p=7680$ km (13) with $T_{ICB}=5500$ K. **B.** Chemical potential as a function of hydrogen concentration at different pressures (330–365 GPa). Solid and dashed lines denote isothermal and adiabatic conditions, respectively. The horizontal dashed line indicates the chemical potential at the ICB. **C.** Radial hydrogen concentration profiles under isothermal and adiabatic conditions.

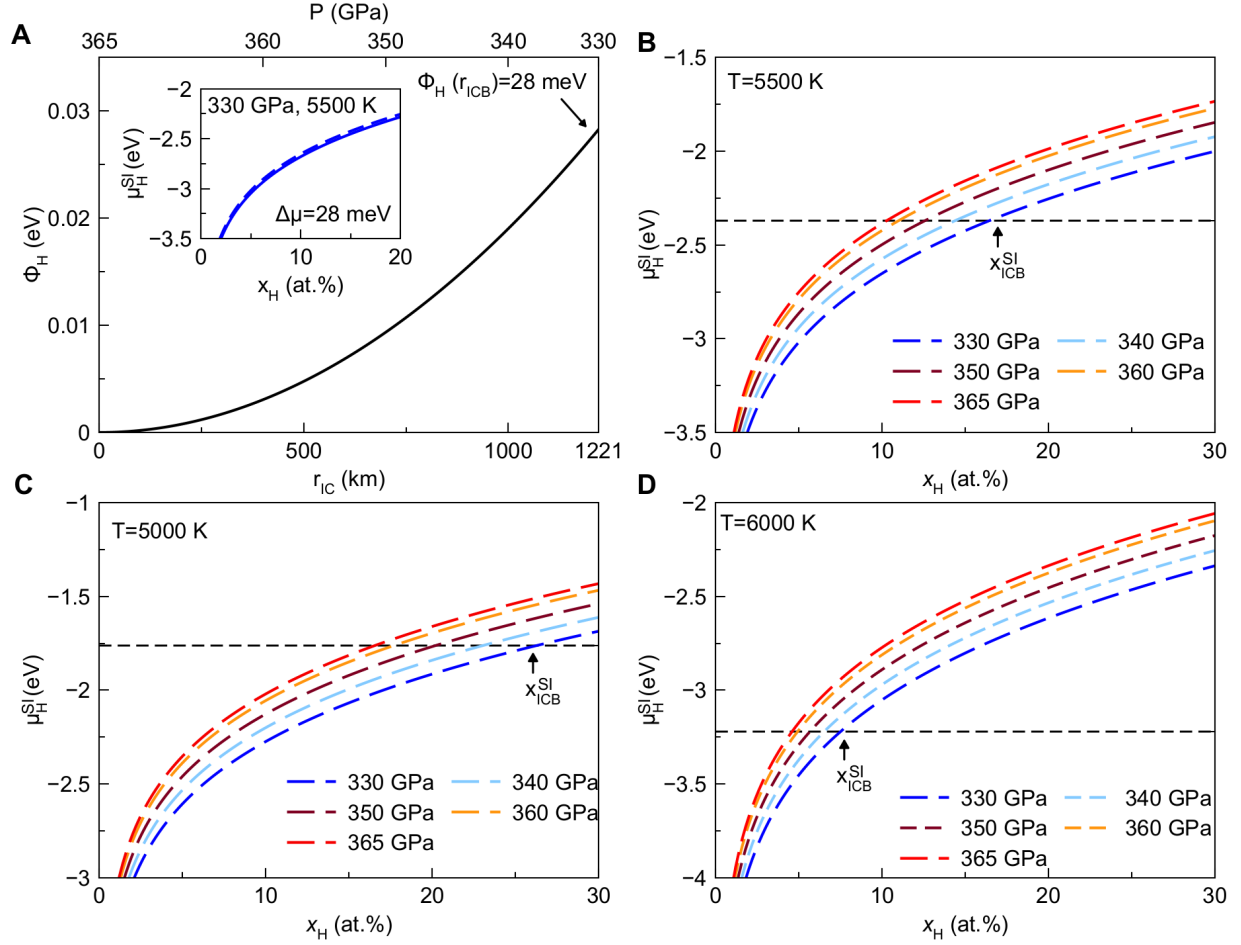


Fig. S18. A. Gravitational potential energy of hydrogen in the inner core as a function of radius (r_{IC}). At the inner core boundary (ICB), $\Phi_H(r_{ICB})$ is ~ 28 meV. This term is added to the chemical potential as a gravitational correction, yielding the total hydrogen chemical potential in a gravitational field. In the inset of panel **A**, the blue solid and dashed lines represent the hydrogen chemical potential at 330 GPa and 5500 K without and with the gravitational correction, respectively. Because pressure maps to radius within the inner core, the gravitational correction varies with pressure. Hydrogen chemical potential versus H content at **B**. 5500 K, **C**. 5000 K, and **D**. 6000 K, including the gravitational correction at pressures corresponding to different inner-core radii (dashed lines).

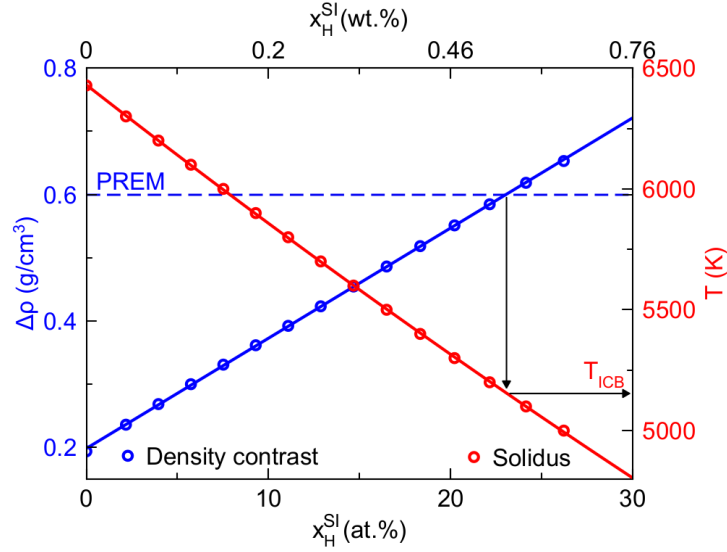


Fig. S19. Density contrast (blue, left axis) between the solid and liquid phases of the $\text{Fe}_{1-x}\text{H}_x$ system, and the Fe-H solidus line (red, right axis) at the ICB pressure. The density contrast is calculated as $\Delta\rho = \rho^S - \rho^L$, where $\rho^S = \frac{(1-x_H^{\text{SI}})m_{\text{Fe}} + x_H^{\text{SI}}m_{\text{H}}}{v^{\text{SI}}(x_H^{\text{SI}})}$ and $\rho^L = \frac{(1-x_H^{\text{L}})m_{\text{Fe}} + x_H^{\text{L}}m_{\text{H}}}{v^{\text{L}}(x_H^{\text{L}})}$. x_H^{SI} and x_H^{L} are taken from the solidus and liquidus lines, respectively, in the phase diagram shown in Fig. 2C. The bottom horizontal axis denotes the mole fraction of superionic H in the solid phase at the ICB, while the *top* horizontal axis shows the corresponding weight fraction. The blue solid line represents the density contrast at ICB inferred from PREM (14).

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